

**PHOTOTHERAPY INDUCED HYPOCALCEMIA IN
TERM AND PRETERM NEONATES WITH
NEONATAL HYPERBILIRUBINEMIA AND ITS
CLINICAL SIGNIFICANCE IN A TERTIARY CARE
HOSPITAL OF SOUTHERN HARYANA**

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AAP	American Academy of Pediatrics
BIND	Bilirubin-Induced Neurological Dysfunction
CBC	Complete Blood Count
DCT	Direct Coombs Test
G6PD	Glucose-6-Phosphate Dehydrogenase
GDM	Gestational Diabetes Mellitus
IV	Intravenous
LBW	Low Birth Weight
LFT	Liver Function Test
LSCS	Lower Segment Cesarean Section
MNCU	Maternal and Neonatal Care Unit
NVD	Normal Vaginal Delivery
NICU	Neonatal Intensive Care Unit
PTH	Parathyroid Hormone
RFT	Renal Function Test

SPSS	Statistical Package for the Social Sciences
TSB	Total Serum Bilirubin
UGT1A1	UDP-Glucuronosyltransferase 1A1
VLBW	Very Low Birth Weight

ABSTRACT

Background

Neonatal hyperbilirubinemia is a common condition requiring prompt management to prevent bilirubin-induced neurological dysfunction. Phototherapy remains the primary treatment modality, but it can induce hypocalcemia, a significant metabolic complication. Despite its clinical relevance, the incidence, risk factors, and outcomes of phototherapy-induced hypocalcemia have not been thoroughly investigated, particularly in term and preterm neonates. This study aimed to evaluate the incidence and risk factors of hypocalcemia in neonates undergoing phototherapy for hyperbilirubinemia and to assess the clinical implications associated with its development.

Methods

This hospital-based prospective observational study was conducted at a tertiary care center in Southern Haryana. A total of 100 neonates requiring phototherapy for hyperbilirubinemia were enrolled. Data on demographic characteristics, birth weight, gestational age, mode of delivery, maternal and neonatal blood groups, and type of phototherapy (conventional or intensive) were collected. Serum bilirubin and calcium levels were measured at the initiation and after 24 hours of phototherapy. The incidence of hypocalcemia was recorded, and its association with various risk factors was analyzed. Statistical analysis was performed using appropriate tests, with a p-value of <0.05 considered significant.

Results

The majority of neonates (58%) presented for phototherapy on the first day of life, and 63% were term neonates. Normal vaginal delivery (53%) was the most common mode of delivery. The mean birth weight was 2.2 kg, with a significant proportion falling into the low birth weight category. Conventional phototherapy was the most frequently used method (71%), while 29% of cases required intensive phototherapy.

Serum bilirubin levels significantly decreased from 11.9 mg/dL to 5.5 mg/dL after 24 hours of phototherapy ($p < 0.001$). However, serum calcium levels also dropped significantly from 8.8 mg/dL to 7.9 mg/dL ($p < 0.05$), indicating a notable incidence of phototherapy-induced

hypocalcemia (42%). Among hypocalcemic neonates, 15% exhibited symptoms such as jitteriness, irritability, and seizures, while the remaining 85% remained asymptomatic.

The study found that preterm neonates had a significantly higher incidence of hypocalcemia (54.8%) compared to term neonates (45.2%) ($p < 0.01$). Intensive phototherapy was significantly associated with hypocalcemia (59.5%) compared to conventional phototherapy (40.5%) ($p < 0.001$). Moreover, symptomatic hypocalcemia was more common among neonates receiving intensive phototherapy, particularly in term infants (85.7%) ($p < 0.01$).

Conclusion

Phototherapy-induced hypocalcemia is a common complication, occurring in 42% of neonates undergoing treatment for hyperbilirubinemia. Intensive phototherapy and low birth weight are significant risk factors. Despite the high incidence of asymptomatic cases, symptomatic hypocalcemia poses a potential risk of severe clinical manifestations. Routine calcium monitoring and appropriate supplementation during phototherapy, particularly in high-risk neonates, are recommended to minimize metabolic complications.

Keywords

Neonatal hyperbilirubinemia, phototherapy-induced hypocalcemia, preterm neonates, calcium homeostasis, maternal vitamin D, neonatal intensive care.

CHAPTER I

INTRODUCTION

Neonatal hyperbilirubinemia is one of the most common conditions affecting newborns, characterized by elevated serum bilirubin levels leading to jaundice. It is estimated to affect approximately 60% of term neonates and up to 80% of preterm neonates in the first week of life (1). While physiological jaundice is typically benign and self-limiting, severe hyperbilirubinemia poses a risk of bilirubin-induced neurological dysfunction (BIND) and kernicterus, which can result in irreversible brain damage (2).

Phototherapy has remained the primary treatment for neonatal hyperbilirubinemia for over six decades and is regarded as a safe and effective modality (3). It works by converting bilirubin into water-soluble isomers that can be excreted without hepatic conjugation. However, despite its efficacy, phototherapy is not free from complications. Among these, phototherapy-induced hypocalcemia has emerged as a significant yet often underdiagnosed metabolic disturbance.

Neonatal hypocalcemia is defined as serum calcium levels below 7 mg/dL in term neonates and below 6 mg/dL in preterm neonates (4). It may present clinically with neuromuscular irritability, jitteriness, apnea, laryngospasm, and seizures, potentially leading to severe neonatal morbidity (5). The pathophysiology of phototherapy-induced hypocalcemia is believed to be linked to suppression of melatonin secretion from the pineal gland, which in turn reduces parathyroid hormone (PTH) levels, leading to decreased calcium resorption from bones (6).

Although phototherapy-induced hypocalcemia is well-documented in research, it remains underrecognized in clinical practice. Studies have shown a significant reduction in serum calcium levels following phototherapy. One study observed a drop in mean serum calcium levels from 8.79 ± 1.44 mg/dL to 8.13 ± 1.67 mg/dL after 48 hours of phototherapy ($p < 0.05$), highlighting its potential clinical significance (1). Another study reported that preterm neonates were more susceptible to phototherapy-induced hypocalcemia than term neonates due to their immature parathyroid function and inadequate calcium homeostasis (7).

Despite this evidence, routine monitoring of calcium levels during phototherapy is not a standard practice in many neonatal intensive care units (NICUs), resulting in a high

prevalence of undiagnosed hypocalcemia (8). Since hypocalcemia can present with non-specific symptoms such as lethargy, jitteriness, and feeding difficulties, it is frequently misdiagnosed or overlooked. If left untreated, it may contribute to long-term neurodevelopmental impairments in neonates (9).

The global burden of neonatal hyperbilirubinemia varies based on genetic, ethnic, and environmental factors. In India, studies have estimated that nearly 70% of neonates admitted to hospitals require phototherapy for hyperbilirubinemia management (3). Given the rising incidence of hyperbilirubinemia and the widespread use of phototherapy, there is an urgent need to investigate and address its metabolic side effects, including hypocalcemia (10).

A study conducted in a tertiary care hospital in Jharkhand found that nearly 40% of neonates who received phototherapy developed hypocalcemia, with a higher incidence in preterm neonates (1). Another study demonstrated that intensive phototherapy caused a greater drop in serum calcium levels compared to conventional phototherapy, suggesting that light intensity may be a contributing factor (6).

Preterm neonates are particularly vulnerable to hypocalcemia due to immature parathyroid function and an inadequate calcium-regulating system. Studies suggest that maternal vitamin D deficiency, intrauterine growth restriction, and perinatal stress further increase the risk of neonatal hypocalcemia (7). Given the high prevalence of neonatal jaundice and the increasing dependence on phototherapy, it is imperative to assess the impact of phototherapy on calcium metabolism to develop evidence-based strategies for early detection and prevention.

The exact mechanism underlying phototherapy-induced hypocalcemia is not fully understood, but the most widely accepted hypothesis involves the inhibition of melatonin secretion from the pineal gland. Melatonin plays a crucial role in calcium homeostasis by stimulating PTH secretion, which facilitates calcium mobilization from bones and reduces urinary calcium excretion (6).

Phototherapy inhibits melatonin production, leading to a decrease in PTH secretion, which subsequently results in reduced calcium resorption from bones and increased renal calcium loss (4). The extent of calcium depletion is influenced by several factors:

- Duration of phototherapy: Prolonged phototherapy increases the risk of hypocalcemia (1).

- Light intensity: Intensive phototherapy has been shown to cause a more significant drop in calcium levels compared to conventional phototherapy (6).
- Gestational age: Preterm neonates have a blunted parathyroid response, making them more susceptible (7).
- Maternal nutritional status: Vitamin D deficiency and inadequate calcium intake during pregnancy have been linked to an increased risk of neonatal hypocalcemia (5).

Despite compelling evidence on the prevalence and risk factors of phototherapy-induced hypocalcemia, routine calcium monitoring is not included in current neonatal care guidelines (8). This gap in clinical practice increases the risk of undetected hypocalcemia, leading to serious neonatal complications. Since hypocalcemia can present with non-specific symptoms that mimic other neonatal conditions such as sepsis or hypoglycemia, it is often misdiagnosed and inappropriately managed (9).

This study is particularly significant in a tertiary care hospital setting in Southern Haryana, where a high volume of neonates receive phototherapy for hyperbilirubinemia. Identifying the prevalence, risk factors, and clinical consequences of hypocalcemia will help in developing targeted interventions such as routine calcium monitoring and supplementation strategies to improve neonatal outcomes.

Moreover, given the high burden of neonatal hyperbilirubinemia in India, findings from this study can inform future guidelines for neonatal care. Understanding the impact of phototherapy on calcium metabolism will help optimize treatment protocols, minimize complications, and improve neonatal survival rates.

Phototherapy remains the gold standard treatment for neonatal hyperbilirubinemia; however, its complications, including hypocalcemia, warrant careful evaluation. Current neonatal care guidelines do not mandate routine calcium monitoring during phototherapy, increasing the risk of undiagnosed and untreated hypocalcemia. This study seeks to bridge this gap by assessing the incidence, risk factors, and clinical implications of phototherapy-induced hypocalcemia in neonates. The findings will help in formulating new guidelines for monitoring and early intervention, ensuring safer phototherapy practices and improved neonatal health outcomes.

CHAPTER II

AIM & OBJECTIVES

To estimate the phototherapy induced hypocalcemia in term and preterm neonates with neonatal hyperbilirubinemia in a tertiary care hospital of southern Haryana.

CHAPTER III

REVIEW OF LITERATURE

Neonatal Hyperbilirubinemia: Pathophysiology, Causes, and Clinical Significance

Neonatal hyperbilirubinemia is a widely recognized condition in neonatology, characterized by an elevation of serum bilirubin levels leading to jaundice. It is one of the most frequently observed clinical findings in newborns and affects approximately 60% of term neonates and more than 80% of preterm neonates during the first week of life (10). While mild hyperbilirubinemia is typically benign and transient, severe cases can lead to bilirubin-induced neurological dysfunction (BIND) and kernicterus, which can result in irreversible neurodevelopmental impairments (11).

The global burden of neonatal hyperbilirubinemia varies based on genetic, environmental, and healthcare factors. A study conducted in Australia reported the incidence of extreme neonatal hyperbilirubinemia (total serum bilirubin $\geq 450 \mu\text{mol/L}$) to be 9.4 per 100,000 live births, identifying ABO incompatibility, glucose-6-phosphate dehydrogenase (G6PD) deficiency, and Rhesus (Rh) isoimmunization as major contributors (12). A Danish study spanning 15 years estimated the incidence of severe neonatal hyperbilirubinemia at 42 per 100,000 neonates, with ABO isohemolytic disease being the most frequent underlying cause (13). In developing countries like India, where neonatal healthcare infrastructure is still evolving, hyperbilirubinemia remains a leading cause of neonatal hospitalization, with approximately 70% of admitted neonates requiring phototherapy (14).

Physiology of Bilirubin Metabolism in Neonates

Bilirubin is a byproduct of hemoglobin breakdown and is primarily eliminated through hepatic metabolism and excretion. The bilirubin metabolic pathway comprises three essential steps: production, hepatic conjugation, and excretion. Hemoglobin from aging erythrocytes undergoes degradation in the reticuloendothelial system, leading to the formation of unconjugated bilirubin, which is lipid-soluble and neurotoxic (15).

The liver plays a crucial role in the metabolism of bilirubin through conjugation with glucuronic acid, a process mediated by UDP-glucuronosyltransferase (UGT1A1). However, neonates, particularly preterm infants, have immature UGT1A1 activity, resulting in an accumulation of unconjugated bilirubin (16). Once conjugated, bilirubin becomes water-soluble and is excreted through the bile into the gastrointestinal tract. However, immature gut flora and increased enterohepatic circulation in neonates contribute to enhanced reabsorption of bilirubin, further increasing serum levels (17).

Emerging studies highlight the genetic regulation of bilirubin metabolism, particularly involving mutations in the UGT1A1 gene, which reduce the liver's ability to conjugate bilirubin. Deficiencies in this enzyme are associated with conditions such as Gilbert's syndrome and Crigler-Najjar syndrome, both of which are linked to persistent neonatal jaundice (18).

Causes of Neonatal Hyperbilirubinemia

Neonatal hyperbilirubinemia can be categorized as either physiological or pathological based on its underlying etiology and clinical presentation.

Physiological jaundice is the most common benign form of neonatal hyperbilirubinemia, observed in nearly two-thirds of term neonates. It arises due to a combination of increased bilirubin production, immature hepatic conjugation, and enhanced enterohepatic circulation. This type of jaundice typically manifests around the second or third day of life, peaks between days four and five, and resolves by 10–14 days without medical intervention (19).

Pathological jaundice, in contrast, occurs within the first 24 hours of life or persists beyond two weeks, often requiring prompt medical attention. Among the leading causes of pathological hyperbilirubinemia, hemolytic diseases are of particular concern. ABO and Rh incompatibility result in immune-mediated hemolysis, wherein maternal antibodies target fetal erythrocytes, leading to increased bilirubin production and severe anemia (20). Another common hemolytic disorder, G6PD deficiency, affects millions of newborns globally and is particularly prevalent in Asian and Mediterranean populations. This enzyme deficiency predisposes neonates to oxidative stress-induced hemolysis, exacerbating bilirubin overload (21).

Apart from hemolytic conditions, non-hemolytic causes such as breast milk jaundice, neonatal infections, and birth trauma significantly contribute to hyperbilirubinemia. Breast milk jaundice, occurring in 2–4% of neonates, is thought to result from breast milk compounds inhibiting UGT1A1 activity, delaying bilirubin conjugation and excretion (23). Neonatal infections, particularly urinary tract infections (UTIs) and congenital TORCH infections, can impair hepatic function, further complicating bilirubin metabolism. Birth trauma, including cephalohematoma and extensive bruising, leads to increased erythrocyte turnover, exacerbating bilirubin production (24).

A retrospective study from China, analyzing 1602 neonates with hyperbilirubinemia, found that hemolysis accounted for 15.17% of cases, breast milk jaundice for 12.09%, infections for 10.17%, and G6PD deficiency for 9.14%, while 41.72% of cases had no identifiable cause (26). These findings underscore the multifactorial nature of neonatal hyperbilirubinemia, necessitating a comprehensive approach to diagnosis and management.

Clinical Implications of Severe Hyperbilirubinemia

Hyperbilirubinemia is typically classified based on total serum bilirubin (TSB) levels into mild, moderate, and severe forms. Mild cases (TSB <12 mg/dL) often resolve without intervention, whereas moderate cases (TSB 12–20 mg/dL) require phototherapy to prevent complications. Severe hyperbilirubinemia (TSB >20 mg/dL) poses a high risk of bilirubin encephalopathy and kernicterus, warranting urgent medical intervention (27).

One of the most serious consequences of untreated neonatal hyperbilirubinemia is bilirubin-induced neurological dysfunction (BIND). When unconjugated bilirubin crosses the blood-brain barrier, it accumulates in specific brain regions, including the basal ganglia and brainstem nuclei, leading to neuronal toxicity.

The clinical spectrum of BIND includes both acute and chronic forms. Acute bilirubin encephalopathy (ABE) presents with lethargy, poor feeding, high-pitched crying, hypotonia, and opisthotonus in the early stages. If left untreated, ABE can progress to seizures, apnea, and coma, with a high risk of mortality. Chronic bilirubin toxicity results in kernicterus, a condition characterized by athetoid cerebral palsy, sensorineural hearing loss, and cognitive impairments (28).

A study in China evaluating 614 neonates requiring exchange transfusion found that 53.5% of infants with ABO incompatibility developed acute bilirubin encephalopathy, and 4.3% suffered long-term neurological sequelae. Similarly, a recent study on bilirubin neurotoxicity identified that elevated unbound bilirubin can lead to apoptosis and neuroinflammation, increasing the risk of autism spectrum disorders (ASD) in affected neonates (29).

Experimental research using a mouse model of bilirubin neurotoxicity demonstrated that pharmacological inhibition of neuroinflammatory pathways significantly improved survival and neurological outcomes, suggesting potential therapeutic avenues for severe neonatal hyperbilirubinemia (30).

Neonatal hyperbilirubinemia remains a significant clinical concern, with a high prevalence worldwide. While physiological jaundice is self-limiting, pathological hyperbilirubinemia resulting from hemolytic diseases, infections, genetic factors, and metabolic disorders requires timely intervention to prevent neurological complications. Given the risk of bilirubin-induced neurotoxicity and kernicterus, early detection and appropriate management strategies, including phototherapy and exchange transfusion, are essential in reducing neonatal morbidity and mortality.

Phototherapy as the Mainstay of Treatment for Neonatal Jaundice

Phototherapy is the primary and most effective treatment for neonatal hyperbilirubinemia. It works by transforming unconjugated bilirubin into water-soluble photoisomers, allowing for excretion through the bile and urine without hepatic conjugation. This therapeutic approach has been in use for over six decades and is considered the first-line treatment for neonatal jaundice due to its efficacy and non-invasive nature. The widespread adoption of phototherapy has significantly reduced the need for exchange transfusion, which was historically the primary treatment for severe neonatal jaundice. However, despite its benefits, phototherapy is not without risks, and concerns regarding its long-term safety, potential genotoxic effects, and metabolic disturbances have gained increasing attention (31).

Mechanism of Phototherapy

Phototherapy reduces bilirubin levels by inducing structural and configurational changes in unconjugated bilirubin molecules. Blue light within the wavelength range of 460–490 nm is most effective in breaking down bilirubin into three major photoisomers:

1. Lumirubin – A water-soluble bilirubin derivative that is rapidly excreted through bile and urine.
2. Configurational Isomers – Bilirubin molecules undergo structural alteration but remain unconjugated.
3. Structural Isomers – Permanent chemical changes that facilitate bilirubin elimination.

The efficiency of phototherapy depends on several factors, including wavelength, intensity, surface area of exposure, and duration. The greater the skin exposure to light, the higher the bilirubin breakdown rate, making double phototherapy (using two light sources) more effective than single phototherapy in severe cases. Studies have shown that LED phototherapy is more effective than conventional fluorescent tube phototherapy due to its narrow-band spectrum and higher irradiance levels (32).

Types of Phototherapy

Phototherapy can be categorized into conventional, intensive, and fiberoptic phototherapy, each varying in terms of light intensity, duration, and mode of administration.

Conventional Phototherapy is the most widely used form, where fluorescent or LED light sources are placed above the neonate in an incubator or open crib. The baby is positioned without clothing except for a diaper and eye protection, ensuring maximum skin exposure.

Intensive Phototherapy delivers higher light intensity ($>30 \mu\text{W}/\text{cm}^2/\text{nm}$) and is typically used in cases where serum bilirubin levels rise rapidly or approach the exchange transfusion threshold. Studies have demonstrated that intensive phototherapy can lower bilirubin levels by 0.5–1 mg/dL per hour, reducing the need for more invasive treatments.

Fiberoptic Phototherapy uses light-emitting fiberoptic blankets placed directly in contact with the neonate's skin, allowing for continuous therapy while enabling maternal bonding. This method is particularly useful for preterm neonates and those at risk of dehydration from excessive phototherapy exposure. Studies suggest that fiberoptic phototherapy is less

effective than conventional overhead phototherapy but can be useful as an adjunctive therapy (33).

Guidelines for Phototherapy Use

The American Academy of Pediatrics (AAP) guidelines provide standardized recommendations for initiating phototherapy in neonates based on their gestational age, total serum bilirubin (TSB) levels, and risk factors. For term infants, phototherapy is initiated if TSB exceeds 15 mg/dL at 24–48 hours, 18 mg/dL at 49–72 hours, and 20 mg/dL after 72 hours. Preterm neonates have lower threshold levels due to immature hepatic function and increased bilirubin vulnerability.

In addition to the AAP guidelines, the National Institute for Health and Care Excellence (NICE) guidelines recommend continuous bilirubin monitoring and individualized treatment plans for neonates requiring prolonged phototherapy. Both guidelines emphasize avoiding unnecessary phototherapy in cases of mild jaundice to minimize potential side effects (34).

Adverse Effects of Phototherapy

Although phototherapy is generally considered safe, emerging research has raised concerns regarding short-term and long-term adverse effects.

One of the most common complications of phototherapy is dehydration. The increased insensible water loss due to continuous light exposure can lead to electrolyte imbalances, hypernatremia, and weight loss, particularly in preterm neonates. Frequent breastfeeding or intravenous fluid supplementation is recommended to counteract this effect.

Another well-documented concern is phototherapy-induced hypocalcemia, which occurs due to suppression of melatonin secretion from the pineal gland, subsequently reducing parathyroid hormone (PTH) levels and calcium mobilization. Studies have found that serum calcium levels significantly drop within 48 hours of phototherapy initiation, increasing the risk of neuromuscular irritability, jitteriness, and seizures.

Phototherapy has also been associated with oxidative stress and DNA damage. A study evaluating DNA damage in term neonates undergoing phototherapy found that exposure to high-intensity light correlated with increased markers of oxidative stress and genotoxicity,

raising concerns regarding the long-term carcinogenic potential of phototherapy. However, further research is required to establish the clinical significance of these findings (35).

Another rare but recognized complication is Bronze Baby Syndrome, which occurs in neonates with cholestatic jaundice undergoing phototherapy. The skin develops a grayish-brown discoloration, believed to result from bilirubin photodegradation products accumulating in the skin and bloodstream. While the condition is benign and reversible, it highlights the importance of careful patient selection for phototherapy.

Recent studies have also explored the immunomodulatory effects of phototherapy, suggesting that light exposure may alter neonatal immune responses. There is evidence that neonates exposed to prolonged phototherapy have a higher risk of developing allergic diseases and immune dysregulation later in life, though the exact mechanisms remain unclear.

In addition to metabolic and immunological concerns, phototherapy may also affect cardiac autonomic control. Research has demonstrated that neonates undergoing phototherapy experience alterations in heart rate variability and autonomic nervous system balance, potentially affecting neonatal cardiovascular adaptation. However, the clinical implications of these findings require further investigation (36).

Phototherapy remains the gold standard treatment for neonatal hyperbilirubinemia, significantly reducing the incidence of severe jaundice and kernicterus. Advances in phototherapy technology, including the development of LED and fiberoptic phototherapy, have improved treatment efficacy while minimizing adverse effects. However, despite its widespread use, concerns regarding potential metabolic disturbances, genotoxic effects, and long-term health implications have led to increasing scrutiny. Ensuring judicious use of phototherapy, monitoring neonates for complications, and adhering to established guidelines are crucial in optimizing treatment outcomes while minimizing risks.

Risk Factors for Phototherapy-Induced Hypocalcemia

Phototherapy is the standard treatment for neonatal hyperbilirubinemia; however, it has been linked to hypocalcemia, a condition characterized by a drop in serum calcium levels below the physiological range. Several studies have demonstrated that neonates undergoing phototherapy, especially preterm and low birth weight neonates, experience significant

reductions in serum calcium due to mechanisms that are not yet fully understood. The leading hypothesis suggests that phototherapy suppresses melatonin secretion, which in turn inhibits parathyroid hormone (PTH) release, resulting in reduced calcium mobilization from bones. The risk of hypocalcemia varies among neonates, depending on factors such as gestational age, birth weight, phototherapy duration and intensity, maternal nutritional status, and perinatal conditions. A deeper understanding of these risk factors is crucial for identifying neonates who require close monitoring during phototherapy to prevent adverse effects (37).

Gestational Age and Birth Weight

One of the most significant risk factors for phototherapy-induced hypocalcemia is gestational age. Preterm neonates, particularly those born before 32 weeks of gestation, are at a higher risk of metabolic disturbances, including hypocalcemia. This vulnerability arises from the immaturity of the parathyroid gland, which leads to inadequate calcium homeostasis and an impaired response to falling calcium levels. Unlike term neonates, who can mobilize calcium from bones and enhance renal calcium reabsorption in response to hypocalcemia, preterm neonates have a blunted parathyroid response, resulting in prolonged and more severe hypocalcemia (38).

Birth weight is another critical factor influencing the development of phototherapy-induced hypocalcemia. Low birth weight (LBW) neonates, particularly very low birth weight (VLBW, <1500 g) and extremely low birth weight (ELBW, <1000 g) neonates, have lower calcium reserves due to limited intrauterine calcium accretion. During fetal development, most calcium is transferred from the mother to the fetus during the third trimester of pregnancy, meaning that preterm neonates, who miss this critical period of calcium transfer, are born with reduced bone mineral content and inadequate calcium stores. As a result, these neonates fail to compensate for phototherapy-induced calcium losses, making them highly susceptible to hypocalcemia (39).

A study evaluating the effect of phototherapy on serum calcium levels in neonates found that hypocalcemia occurred in 62% of preterm neonates (<32 weeks gestation) and 40% of term neonates receiving phototherapy. Additionally, VLBW neonates exhibited a more significant drop in calcium levels compared to neonates with birth weights above 2500 g. These findings highlight the critical interplay between gestational age, birth weight, and calcium metabolism, necessitating closer monitoring of high-risk neonates undergoing phototherapy (40).

Duration and Intensity of Phototherapy

The extent of light exposure, in terms of both duration and intensity, plays a crucial role in determining the degree of phototherapy-induced hypocalcemia. Studies have consistently shown that prolonged phototherapy (exceeding 48 hours) leads to a sustained suppression of melatonin secretion, thereby reducing parathyroid hormone release and impairing calcium homeostasis. Neonates who undergo longer durations of phototherapy are more likely to develop severe hypocalcemia compared to those receiving shorter exposure.

The intensity of phototherapy also contributes to calcium depletion. Intensive phototherapy, defined as light exposure with an irradiance of $\geq 30 \mu\text{W}/\text{cm}^2/\text{nm}$, is known to accelerate bilirubin breakdown but also exacerbates the suppression of melatonin secretion, leading to more profound hypocalcemia. A comparative study analyzing intensive vs. conventional phototherapy demonstrated that neonates receiving intensive phototherapy experienced a mean reduction in serum calcium of 1.4 mg/dL, whereas those on conventional phototherapy exhibited a mean reduction of 0.7 mg/dL. These findings suggest that high-intensity phototherapy contributes to more severe calcium depletion and should be cautiously used in neonates at risk of hypocalcemia (41).

Another concern is the effect of multiple phototherapy cycles on calcium metabolism. Neonates who require multiple sessions of phototherapy due to recurrent or refractory hyperbilirubinemia are exposed to repeated cycles of melatonin suppression, leading to cumulative calcium loss. This phenomenon is particularly relevant in preterm neonates, who often experience prolonged jaundice and require extended phototherapy exposure.

Maternal Nutritional Status and Neonatal Calcium Stores

Maternal nutritional status plays a significant role in neonatal calcium homeostasis, as fetal calcium accretion is highly dependent on maternal calcium and vitamin D levels during pregnancy. Maternal vitamin D deficiency is a well-documented risk factor for neonatal hypocalcemia, and its impact is further exacerbated when neonates are subjected to phototherapy.

A study assessing the relationship between maternal vitamin D levels and neonatal calcium status found that neonates born to mothers with vitamin D levels $< 20 \text{ ng/mL}$ had a 2.5-fold increased risk of developing hypocalcemia following phototherapy. The severity of

hypocalcemia was more pronounced in neonates born to mothers with severe vitamin D deficiency (<10 ng/mL), underscoring the importance of adequate maternal vitamin D supplementation during pregnancy (42).

Additionally, maternal parathyroid disorders, chronic kidney disease, and gestational diabetes mellitus (GDM) have been linked to altered fetal calcium metabolism. Neonates born to mothers with hypoparathyroidism or chronic renal disease often have impaired calcium regulation, making them more vulnerable to phototherapy-induced hypocalcemia. Similarly, maternal GDM is associated with an increased risk of neonatal hypocalcemia, likely due to fetal hyperinsulinism suppressing parathyroid function.

Perinatal Conditions and Neonatal Hypocalcemia

Neonates with perinatal stress, birth asphyxia, or sepsis have an increased risk of hypocalcemia when exposed to phototherapy. Birth asphyxia leads to metabolic acidosis, which impairs parathyroid gland function and reduces calcium mobilization from bones, thereby exacerbating the effects of phototherapy on calcium levels. A study comparing neonates with and without birth asphyxia undergoing phototherapy found that 65% of asphyxiated neonates developed hypocalcemia, compared to 30% of non-asphyxiated neonates (43).

Neonatal sepsis is another contributing factor. Infection-induced inflammation alters calcium metabolism by increasing pro-inflammatory cytokine levels, which interfere with PTH secretion and decrease intestinal calcium absorption. Neonates with sepsis who undergo phototherapy exhibit significantly higher rates of hypocalcemia compared to non-septic neonates, suggesting a synergistic effect of infection and phototherapy on calcium depletion.

Meta-Analysis of Risk Factors

A meta-analysis involving over 10,000 neonates receiving phototherapy stratified risk factors for phototherapy-induced hypocalcemia as follows:

- Preterm neonates (<32 weeks) – 70% incidence
- VLBW neonates (<1500 g) – 65% incidence
- Maternal vitamin D deficiency (<20 ng/mL) – 55% incidence
- Phototherapy duration >48 hours – 50% incidence

- Intensive phototherapy ($\geq 30 \mu\text{W}/\text{cm}^2/\text{nm}$) – 45% incidence
- Neonates with perinatal asphyxia or sepsis – 40% incidence

These findings emphasize the need for routine calcium monitoring and supplementation in high-risk neonates undergoing phototherapy (44).

Phototherapy-induced hypocalcemia is a multifactorial condition influenced by gestational age, birth weight, phototherapy exposure, maternal nutritional status, and perinatal complications. Preterm and low birth weight neonates are particularly vulnerable, with the severity of hypocalcemia increasing with longer and more intense phototherapy exposure. Given the potential neurological and metabolic consequences of hypocalcemia, routine serum calcium monitoring, targeted supplementation, and individualized phototherapy protocols are essential to ensuring safe neonatal care.

Clinical Implications and Management of Phototherapy-Induced Hypocalcemia

Phototherapy is widely regarded as a safe and effective treatment for neonatal hyperbilirubinemia, but emerging evidence suggests that it can induce significant metabolic disturbances, particularly hypocalcemia. The clinical implications of hypocalcemia in neonates range from subtle neuromuscular symptoms to life-threatening cardiac and neurological complications. Given that calcium plays a crucial role in cell signaling, enzymatic activity, nerve conduction, and muscle contraction, a disruption in calcium homeostasis can lead to systemic dysfunction in neonates undergoing phototherapy (45).

Despite its clinical significance, phototherapy-induced hypocalcemia remains underdiagnosed due to the nonspecific nature of its symptoms and the lack of routine calcium monitoring in neonatal care protocols. However, studies have demonstrated a statistically significant reduction in serum calcium levels in neonates undergoing phototherapy, with preterm and low birth weight neonates being at the highest risk.

Clinical Manifestations of Phototherapy-Induced Hypocalcemia

The clinical presentation of phototherapy-induced hypocalcemia can be variable, with some neonates remaining asymptomatic, while others develop pronounced neurological,

respiratory, and cardiovascular symptoms. The severity of symptoms often correlates with the degree of calcium depletion, the duration of phototherapy, and the presence of predisposing risk factors such as prematurity and low birth weight.

Neuromuscular Symptoms

One of the most common early manifestations of hypocalcemia in neonates is neuromuscular irritability, which includes jitteriness, hypertonia, exaggerated startle reflexes, and tremors. These symptoms result from increased neuronal excitability due to reduced calcium availability. In severe cases, neonates may exhibit tetanic spasms, laryngospasm, or stridor, requiring urgent calcium correction (46).

Respiratory Complications

Phototherapy-induced hypocalcemia can also impact the respiratory system, particularly in preterm neonates with immature respiratory control. Episodes of apnea and periodic breathing abnormalities have been reported in neonates undergoing phototherapy, with hypocalcemia being a contributing factor. Calcium plays a role in neuromuscular function, and its depletion may impair diaphragmatic contraction, leading to intermittent episodes of respiratory distress (47).

Seizures and Neurotoxicity

Seizures are among the most severe complications of hypocalcemia in neonates. Unlike hypoxic-ischemic encephalopathy (HIE)-related seizures, hypocalcemic seizures are often subtle, presenting as lip smacking, eye deviation, or brief focal clonic movements. These seizures typically respond rapidly to intravenous calcium administration, reinforcing the importance of early recognition and treatment (48).

Cardiac Manifestations

Prolonged hypocalcemia can impact cardiac function, leading to arrhythmias, bradycardia, and prolonged QT interval. Electrocardiographic (ECG) abnormalities have been documented in neonates with severe calcium depletion, with prolonged QTc intervals being a common finding. If left untreated, prolonged QT syndrome can increase the risk of fatal arrhythmias, emphasizing the need for early detection and correction.

A study evaluating the incidence of cardiac dysfunction in neonates with phototherapy-induced hypocalcemia found that 18% of affected neonates developed bradycardia, while 11% exhibited prolonged QT intervals. These findings highlight the

potential cardiac risks associated with phototherapy and the need for routine cardiovascular monitoring (49).

Pathophysiology and Biochemical Changes in Phototherapy-Induced Hypocalcemia

The exact mechanism behind phototherapy-induced hypocalcemia remains under investigation, but the most widely accepted hypothesis involves melatonin suppression and its downstream effects on parathyroid hormone (PTH) secretion.

Melatonin, produced by the pineal gland, plays a key role in regulating calcium homeostasis. Under normal physiological conditions, melatonin stimulates PTH release, which in turn enhances calcium resorption from bones and increases renal calcium reabsorption. Phototherapy, particularly blue light exposure, suppresses melatonin secretion, leading to a reduction in PTH levels and decreased serum calcium concentration (50).

Biochemical changes in neonates with phototherapy-induced hypocalcemia include:

- Decrease in total and ionized calcium levels – Defined as total calcium <7 mg/dL in term neonates and <6 mg/dL in preterm neonates, with ionized calcium levels <4.4 mg/dL.
- Suppressed parathyroid hormone (PTH) levels – Confirming melatonin-mediated inhibition of PTH secretion.
- Elevated phosphorus levels – Due to the inverse relationship between calcium and phosphate homeostasis.
- Altered magnesium levels – Magnesium is a cofactor for PTH secretion, and its depletion can further exacerbate hypocalcemia (51).

Diagnosis and Screening Guidelines for Phototherapy-Induced Hypocalcemia

Currently, there is no universal guideline recommending routine calcium monitoring in neonates receiving phototherapy. However, increasing evidence suggests that routine serum calcium assessments should be incorporated into neonatal care, particularly for high-risk neonates.

A proposed screening protocol includes:

1. Baseline Serum Calcium Measurement – Before initiating phototherapy, particularly in preterm, LBW, and critically ill neonates.
2. Follow-Up Testing at 24 and 48 Hours – To assess the impact of phototherapy on calcium levels and detect early metabolic disturbances.
3. Symptom-Based Testing – Any neonate exhibiting jitteriness, seizures, apnea, or cardiac abnormalities should undergo immediate serum calcium evaluation (51).

A study implementing routine calcium monitoring in neonates undergoing phototherapy found that early detection and calcium supplementation reduced the incidence of symptomatic hypocalcemia by 50%, highlighting the benefits of proactive screening strategies.

Management Strategies for Phototherapy-Induced Hypocalcemia

The management of phototherapy-induced hypocalcemia focuses on preventing complications through early supplementation and close monitoring. Treatment approaches include:

- Oral Calcium Supplementation – Mild cases of hypocalcemia can be managed with oral calcium gluconate (50–75 mg/kg/day), divided into multiple doses. Oral supplementation is particularly effective in preterm neonates requiring long-term calcium support.
- Intravenous Calcium Therapy – Severe hypocalcemia or symptomatic neonates require IV calcium gluconate (10% solution, 1–2 mL/kg over 10 minutes), followed by a continuous calcium infusion if necessary.
- Magnesium Supplementation – Since magnesium deficiency can suppress PTH secretion, neonates with concurrent hypocalcemia and hypomagnesemia should receive IV magnesium sulfate (25–50 mg/kg/day).
- Optimizing Phototherapy Practices – Avoiding unnecessary prolonged exposure to phototherapy, ensuring adequate hydration, and regularly monitoring calcium levels can help reduce the incidence and severity of hypocalcemia (52).

Long-Term Implications and Follow-Up Recommendations

Long-term studies suggest that neonates with prolonged or recurrent hypocalcemia may have an increased risk of neurodevelopmental impairment and bone mineralization deficits. A

follow-up study of neonates diagnosed with phototherapy-induced hypocalcemia found that at 12 months of age, 18% exhibited delayed motor milestones, compared to 6% in the control group. Additionally, preterm neonates with persistent hypocalcemia had lower bone mineral density at 6 months (53).

Recommendations for follow-up include:

1. Neurodevelopmental assessments at 6 and 12 months to monitor motor and cognitive outcomes.
2. Bone mineral density evaluations in preterm neonates with recurrent hypocalcemia.
3. Vitamin D and calcium supplementation beyond the neonatal period, particularly for at-risk neonates.

Phototherapy-induced hypocalcemia is a clinically significant complication that can lead to neurological, respiratory, and cardiac disturbances in neonates. Given its high prevalence in preterm and low birth weight neonates, routine calcium monitoring, early supplementation, and optimized phototherapy practices are essential to ensure safe neonatal management.

Role of Calcium Supplementation in Preventing Phototherapy-Induced Hypocalcemia

Phototherapy is a widely accepted treatment for neonatal hyperbilirubinemia, but its impact on calcium metabolism has raised concerns among neonatologists. Phototherapy-induced hypocalcemia results from melatonin suppression, leading to a decrease in parathyroid hormone (PTH) secretion and impaired calcium mobilization. This phenomenon is particularly prominent in preterm and low birth weight neonates, who already have a limited ability to regulate calcium levels due to immature parathyroid function. Given the potential neuromuscular and cardiovascular complications of hypocalcemia, calcium supplementation has been identified as a key preventive strategy in neonates undergoing phototherapy (54).

Mechanisms Underlying Calcium Depletion and Need for Supplementation

Calcium homeostasis in neonates is primarily regulated by PTH, vitamin D, and renal function. Under normal conditions, PTH enhances calcium resorption from bones and increases renal calcium reabsorption to maintain serum calcium levels. However,

phototherapy disrupts this balance by suppressing melatonin secretion, leading to reduced PTH release, decreased bone calcium resorption, and increased renal calcium excretion. This cascade of events results in hypocalcemia, which can lead to neuromuscular irritability, seizures, apnea, and cardiac dysfunction if left untreated.

Studies have demonstrated that serum calcium levels drop significantly within 24–48 hours of phototherapy initiation, with neonates in intensive phototherapy exhibiting greater reductions compared to those receiving conventional phototherapy. This highlights the need for proactive calcium supplementation to maintain stable calcium levels and prevent severe complications (55).

Comparing Oral and Intravenous Calcium Supplementation

Calcium supplementation is primarily administered via two routes: oral and intravenous. The choice between these depends on the severity of hypocalcemia, clinical presentation, and overall neonatal condition.

Oral calcium supplementation is generally recommended for mild-to-moderate hypocalcemia or prophylactic use. Calcium gluconate (50–75 mg/kg/day in divided doses) is the preferred formulation due to its high bioavailability and favorable safety profile. Oral administration is less invasive than intravenous therapy and is particularly beneficial for neonates who do not exhibit acute symptoms. A study comparing neonates receiving prophylactic oral calcium with those who did not found that the incidence of symptomatic hypocalcemia was significantly lower (18% vs. 45%) in the supplementation group, emphasizing its effectiveness in preventing complications (56).

Intravenous (IV) calcium supplementation, on the other hand, is reserved for severe hypocalcemia or symptomatic cases, such as neonates experiencing seizures, apnea, or cardiac arrhythmias. The standard treatment involves administering IV calcium gluconate 10% (1–2 mL/kg over 10 minutes), followed by continuous infusion if necessary. IV calcium rapidly restores calcium levels but requires careful monitoring due to risks such as extravasation, bradycardia, and hypercalcemia. A clinical trial comparing IV vs. oral calcium supplementation found that while IV therapy corrected hypocalcemia faster, oral supplementation was equally effective in maintaining normal calcium levels over time, making it the preferred option for stable neonates (57).

Optimal Timing and Duration of Calcium Supplementation

The timing of calcium supplementation plays a crucial role in its effectiveness. Prophylactic administration, initiated before or within the first 12 hours of phototherapy, has been shown to reduce the incidence of severe hypocalcemia by up to 40%. This early intervention prevents sudden calcium depletion and mitigates the risk of symptomatic hypocalcemia.

The duration of supplementation varies based on neonatal response, but studies suggest continuing calcium therapy for at least 48–72 hours post-phototherapy to prevent rebound hypocalcemia. Routine serum calcium monitoring every 12–24 hours ensures optimal dosing adjustments, preventing both under-treatment and overcorrection. A longitudinal study reported that neonates who received early calcium supplementation had significantly lower rates of seizures and cardiac disturbances compared to those who received supplementation only after developing hypocalcemia (58).

Long-Term Effects of Calcium Supplementation on Neonatal Health

While the immediate benefits of calcium supplementation in preventing phototherapy-induced hypocalcemia are well documented, researchers are increasingly focusing on its long-term impact on neonatal development, skeletal growth, and metabolic health.

Neurodevelopmental outcomes are an area of particular concern. Hypocalcemia-induced neuronal excitability can interfere with cognitive development, leading to delayed motor milestones and impaired learning abilities. A follow-up study of neonates who experienced severe, untreated hypocalcemia found that at 12 months of age, 30% exhibited neurodevelopmental delays, compared to only 12% in neonates who received early calcium supplementation (59).

Another critical aspect is bone mineralization. Calcium plays a fundamental role in skeletal development, and prolonged hypocalcemia can impair bone growth and reduce bone mineral density. A study assessing preterm neonates with recurrent hypocalcemia found that they had lower bone mineral density at 6 months of age, reinforcing the need for adequate calcium intake in the early neonatal period.

In terms of metabolic and endocrine health, neonatal calcium imbalances may have long-term effects on parathyroid function and metabolic bone disease. Research suggests that neonates who experience repeated episodes of hypocalcemia are at an increased risk of metabolic bone disorders in later childhood, highlighting the importance of early correction and long-term follow-up (60).

Challenges and Risks of Calcium Supplementation

Despite its well-documented benefits, calcium supplementation must be carefully managed to avoid complications.

One of the primary risks associated with IV calcium therapy is extravasation, which can cause severe tissue necrosis if calcium solutions infiltrate surrounding tissues. Strict monitoring of IV administration sites and slow infusion rates can help mitigate this risk.

Another concern is hypercalcemia, which can result from overcorrection or excessive supplementation. Neonates with hypercalcemia may present with lethargy, vomiting, poor feeding, and renal dysfunction. Regular serum calcium monitoring is essential to prevent excessive calcium accumulation (61).

Additionally, calcium homeostasis is closely linked to magnesium and phosphorus levels. Hypomagnesemia impairs PTH secretion, further exacerbating hypocalcemia. Studies have found that neonates receiving both calcium and magnesium supplementation had more stable serum calcium levels compared to those receiving calcium alone, supporting the need for comprehensive electrolyte monitoring.

Calcium supplementation is a proven and effective strategy for preventing phototherapy-induced hypocalcemia. Oral calcium supplementation is preferred for prophylaxis and mild cases, whereas IV calcium is reserved for severe or symptomatic hypocalcemia. The timing and duration of supplementation are critical, with early intervention reducing the incidence of complications. Long-term follow-up is essential to assess neurodevelopment, skeletal health, and metabolic stability in neonates affected by early-life calcium imbalances. By optimizing calcium supplementation protocols, neonatologists can improve neonatal outcomes and reduce the risk of hypocalcemia-related morbidity.

Alternative and Adjunctive Therapies for Managing Neonatal Hyperbilirubinemia

While phototherapy remains the gold standard for treating neonatal hyperbilirubinemia, emerging research has explored alternative and adjunctive therapies that may enhance treatment outcomes, reduce bilirubin levels more efficiently, and minimize the adverse effects associated with prolonged phototherapy. These alternatives include pharmacological agents, probiotics, exchange transfusion, intravenous immunoglobulin (IVIG), and herbal therapies. Some of these approaches are particularly useful in neonates with severe hyperbilirubinemia, hemolytic disease, or those at risk of kernicterus.

Pharmacological Agents in the Management of Neonatal Hyperbilirubinemia

Several drugs have been studied as potential adjuncts to phototherapy to accelerate bilirubin metabolism and reduce the need for exchange transfusion. The most commonly investigated pharmacological agents include:

Phenobarbital

Phenobarbital is a cytochrome P450 enzyme inducer that stimulates hepatic glucuronyl transferase (UGT1A1) activity, enhancing the conjugation and excretion of bilirubin. It has been found to be particularly useful in preterm neonates with immature hepatic enzyme function. A clinical trial comparing phenobarbital plus phototherapy vs. phototherapy alone found that neonates receiving phenobarbital had a faster decline in total serum bilirubin (TSB) levels and required shorter durations of phototherapy (62). However, concerns about sedative effects and neurodevelopmental impact have limited its routine use.

Clofibrate

Clofibrate, a peroxisome proliferator-activated receptor (PPAR- α) agonist, enhances bilirubin conjugation and excretion. Studies suggest that a single oral dose of clofibrate (50 mg/kg) significantly reduces TSB levels in neonates undergoing phototherapy, reducing both the duration of phototherapy and the risk of requiring exchange transfusion (63). Unlike phenobarbital, clofibrate has minimal sedative effects, making it a safer alternative for hyperbilirubinemia management.(96)

Zinc Supplements

Zinc inhibits the enterohepatic circulation of bilirubin by blocking beta-glucuronidase activity in the intestines, thereby reducing bilirubin reabsorption. Some studies have reported that oral zinc supplementation (10 mg/day) in neonates with hyperbilirubinemia leads to a significant reduction in TSB levels and phototherapy duration (64). However, conflicting results exist, and zinc supplementation is not yet part of standard neonatal jaundice management.

Role of Probiotics in Neonatal Jaundice Management

Probiotics have gained attention in recent years due to their role in gut microbiome modulation and bilirubin metabolism. The neonatal intestine lacks a mature microbiota, leading to enhanced enterohepatic circulation of bilirubin. Probiotics, particularly *Lactobacillus* and *Bifidobacterium* species, help establish beneficial gut flora, reduce bilirubin reabsorption, and promote its excretion.(97)

A meta-analysis of randomized controlled trials found that neonates receiving probiotic supplementation had lower bilirubin levels and shorter phototherapy durations compared to controls. The beneficial effects of probiotics are likely due to:

- Inhibition of beta-glucuronidase activity, reducing bilirubin reabsorption.
- Enhancement of gut motility and fecal bilirubin excretion.
- Modulation of inflammatory pathways, improving overall neonatal gut health.

Despite promising results, further large-scale studies are needed before probiotics can be routinely recommended in neonatal jaundice management (65).

Intravenous Immunoglobulin (IVIG) Therapy in Hemolytic Hyperbilirubinemia

Neonates with immune-mediated hemolysis, such as Rh incompatibility and ABO hemolytic disease, are at high risk of developing severe hyperbilirubinemia requiring exchange transfusion. IVIG has been proposed as an effective adjunct therapy in these cases by blocking Fc receptor-mediated hemolysis of sensitized red blood cells, thereby reducing bilirubin production.(93)

A randomized controlled trial found that neonates with hemolytic jaundice receiving IVIG (0.5–1 g/kg over 2 hours) had significantly lower TSB levels and a reduced need for exchange transfusion compared to those receiving phototherapy alone (66). Current

guidelines recommend IVIG in severe isoimmune hemolysis, particularly in cases where TSB levels approach exchange transfusion thresholds.

Exchange Transfusion for Severe or Refractory Hyperbilirubinemia

Despite advancements in phototherapy and pharmacological interventions, some neonates still develop dangerously high bilirubin levels, necessitating exchange transfusion. This procedure involves removing bilirubin-laden blood and replacing it with donor blood, rapidly lowering TSB levels and preventing bilirubin-induced neurotoxicity (kernicterus).(94)

Although exchange transfusion is highly effective, it is associated with risks such as volume overload, electrolyte imbalances, thrombocytopenia, and infection. Advances in phototherapy and IVIG therapy have significantly reduced the need for exchange transfusion, but it remains a lifesaving option in extreme cases (67).

Although phototherapy remains the first-line treatment for neonatal jaundice, alternative and adjunctive therapies are being increasingly explored to enhance treatment efficacy, reduce bilirubin levels more efficiently, and minimize the risks associated with prolonged phototherapy or exchange transfusion. Pharmacological agents such as phenobarbital, clofibrate, and zinc have shown promise in accelerating bilirubin metabolism, while probiotics and IVIG have demonstrated efficacy in specific clinical scenarios. Exchange transfusion remains a lifesaving measure for neonates with severe hyperbilirubinemia refractory to other treatments, while herbal therapies are still under investigation. Further large-scale clinical trials are needed to determine the safety, efficacy, and standardization of these emerging therapies.(95)

EARLIER STUDIES

Vitamin D and Its Role in Neonatal Hypocalcemia

A study by Stessman and Peebles (2018) explored the role of vitamin D in neonatal neurological development and its association with neonatal metabolic disorders, including hypocalcemia. The researchers highlighted that neonates are at an increased risk of vitamin D deficiency due to maternal insufficiency during pregnancy, which can impair calcium homeostasis and increase susceptibility to metabolic disturbances, including phototherapy-induced hypocalcemia. The study further emphasized that vitamin D plays a critical role in parathyroid hormone (PTH) regulation, which is necessary for calcium mobilization in neonates. Although the research primarily focused on neonatal hypoxic-ischemic brain injury, it provided substantial evidence linking low vitamin D levels with impaired neonatal calcium metabolism, suggesting that vitamin D supplementation may reduce the risk of hypocalcemia in neonates requiring phototherapy (68).

Vitamin D Deficiency and Neonatal Outcomes

Mansur et al. (2022) examined the effects of vitamin D deficiency before, during, and after pregnancy on neonatal health, focusing on its implications for metabolic disorders. The study established that low maternal vitamin D levels contribute to neonatal hypocalcemia and increase the risk of late hypocalcemia following phototherapy exposure. Beyond its classic function in bone health and calcium regulation, the study linked vitamin D deficiency to low birth weight, preterm delivery, and increased susceptibility to metabolic disturbances. The authors emphasized that maternal vitamin D supplementation during pregnancy significantly reduces neonatal hypocalcemia, a finding relevant to preventing phototherapy-induced calcium depletion (69).

Vitamin D Requirements During Pregnancy and Neonatal Hypocalcemia Prevention

Kiely et al. (2017) reviewed the current recommendations for maternal vitamin D intake during pregnancy and its impact on neonatal calcium homeostasis. The authors noted that neonatal vitamin D status is directly determined by maternal circulating 25-hydroxyvitamin D concentrations, reinforcing the importance of adequate maternal vitamin D intake in preventing neonatal hypocalcemia. The study found that regulatory guidelines for vitamin D

intake in pregnancy remain insufficient, as they are based on non-pregnant adult requirements. The authors recommended that neonates born to vitamin D-deficient mothers should be closely monitored for hypocalcemia, particularly if they require phototherapy, as these neonates are at a higher risk of calcium depletion (70).

Vitamin A Deficiency and Its Impact on Neonatal Calcium Regulation

Ishaq et al. (2023) investigated the effects of vitamin A imbalance on maternal and neonatal health, noting that vitamin A deficiency during pregnancy can contribute to metabolic disorders, including calcium imbalances. While the study primarily focused on maternal nutrition and fetal immune function, it highlighted that excessive vitamin A intake could lead to hypercalcemia, whereas deficiency was associated with impaired calcium regulation. Although vitamin A itself is not directly linked to phototherapy-induced hypocalcemia, its interaction with vitamin D and calcium metabolism suggests that maintaining balanced vitamin A levels is necessary for neonatal calcium homeostasis. These findings reinforce the need for a comprehensive maternal nutritional approach to prevent neonatal metabolic disturbances (71).

Maternal Vitamin D Deficiency and Neonatal Hypocalcemia Risk

Zhang et al. (2022) explored the relationship between maternal vitamin D levels and neonatal health outcomes, finding a strong association between maternal vitamin D deficiency and neonatal hypocalcemia. The study noted that low maternal vitamin D status increased the risk of poor fetal skeletal growth, intrauterine growth restriction, and neonatal hypocalcemia. The authors proposed that maternal vitamin D supplementation during pregnancy could reduce the risk of neonatal calcium disturbances, supporting the need for prenatal nutritional interventions. Given the established link between maternal vitamin D levels and neonatal calcium regulation, this study provides strong evidence that improving maternal vitamin D status may reduce phototherapy-induced hypocalcemia in neonates (72).

Home Phototherapy for Neonatal Hyperbilirubinemia and Risk of Hypocalcemia

A study by Chang et al. (2020) investigated the use of home-based phototherapy for neonatal jaundice and its potential risks, including hypocalcemia and hospital readmission. The study followed 1,324 neonates born at ≥ 35 weeks gestation who underwent home phototherapy, with daily monitoring of bilirubin and metabolic parameters. Although home phototherapy

successfully reduced bilirubin levels in most cases, 1.9% of neonates required hospital readmission, with hypocalcemia cited as one of the metabolic complications leading to further medical intervention. This study provides valuable insights into the need for routine calcium monitoring, even in outpatient phototherapy settings, to prevent undiagnosed hypocalcemia (73).

Risk of Autism and Long-Term Neurological Effects of Phototherapy-Induced Metabolic Disturbances

Wu et al. (2016) conducted a retrospective cohort study involving 525,409 infants to explore whether phototherapy-induced metabolic alterations, including hypocalcemia, could have long-term neurodevelopmental consequences such as autism spectrum disorder (ASD). While the study initially found an association between severe neonatal hyperbilirubinemia, phototherapy, and increased ASD risk, after adjusting for confounding factors, the link between phototherapy and ASD was no longer statistically significant. However, the study reinforced that birth weight <2500 g and prolonged phototherapy exposure were independent risk factors for metabolic disturbances, including hypocalcemia and its potential effects on neurodevelopment. These findings underscore the importance of early metabolic screening in neonates undergoing phototherapy (74).

Umbilical Cord Blood Bilirubin, Gestational Age, and Predicting Phototherapy Needs

Castillo et al. (2018) developed a predictive model using umbilical cord blood bilirubin (CBB) levels, gestational age, and maternal factors to determine which neonates would require early phototherapy and metabolic monitoring. The study found that elevated CBB levels were strongly correlated with severe hyperbilirubinemia and an increased need for phototherapy. The predictive accuracy of CBB was further improved when combined with gestational age and maternal race, suggesting that preterm neonates and those with high CBB levels are at a greater risk of bilirubin-related complications, including hypocalcemia. This study highlights the importance of early metabolic risk assessment, which can help clinicians identify neonates who may require closer calcium monitoring during phototherapy (75).

Effects of Phototherapy on Serum Calcium and Magnesium Levels in Neonates

Murmu et al. (2024) conducted a prospective observational study on 221 neonates undergoing phototherapy, analyzing changes in serum calcium and magnesium levels before and after treatment. The study found that mean serum calcium levels dropped significantly from 8.79 ± 1.44 mg/dL to 8.13 ± 1.67 mg/dL post-phototherapy, confirming that phototherapy-induced calcium depletion is a clinically significant issue. Additionally, magnesium levels also declined, suggesting that phototherapy may contribute to broader metabolic imbalances beyond calcium homeostasis. These findings support the routine monitoring of calcium and magnesium levels in neonates receiving phototherapy and raise questions about the potential need for combined calcium-magnesium supplementation strategies (1).

Impact of Phototherapy on Breastfeeding Duration and Exclusivity

Waite et al. (2016) examined whether neonatal phototherapy affects breastfeeding duration and exclusivity, considering its potential effects on metabolic regulation, including hypocalcemia. The study analyzed data from 4,441 infants, of which 220 (5%) received phototherapy. While phototherapy did not substantially impact overall breastfeeding rates in the first year, it was associated with lower rates of exclusive breastfeeding in the first four months. The authors suggested that neonates undergoing phototherapy may experience early feeding difficulties, potentially linked to subclinical hypocalcemia, which can cause poor feeding and increased irritability. These findings highlight the need for comprehensive neonatal nutritional support during phototherapy to mitigate calcium-related feeding issues (76).

Phototherapy-Induced Changes in Serum Calcium and Magnesium Levels

Murmu et al. (2024) conducted a prospective observational study analyzing the effects of phototherapy on serum calcium and magnesium levels in neonates with hyperbilirubinemia. The study included 221 neonates and found that mean serum calcium levels significantly declined from 8.79 ± 1.44 mg/dL to 8.13 ± 1.67 mg/dL after 48 hours of phototherapy. Similarly, serum magnesium levels also decreased significantly, suggesting that phototherapy-induced metabolic changes extend beyond calcium depletion. The study concluded that neonates undergoing phototherapy should be routinely monitored for calcium and magnesium imbalances, particularly those requiring prolonged treatment. These findings

reinforce the importance of combined calcium-magnesium supplementation strategies in neonates at risk of phototherapy-induced metabolic disturbances (1).

Complications of Phototherapy and Strategies for Reducing Exposure Duration

Shoris et al. (2023) reviewed the short- and long-term complications of neonatal phototherapy, emphasizing the need to optimize exposure duration to minimize metabolic disturbances such as hypocalcemia. The study highlighted that phototherapy is generally safe but may lead to unintended metabolic effects, including oxidative stress, electrolyte imbalances, and transient immunological alterations. The authors explored various complementary treatments, such as enteral calcium supplementation and modified phototherapy protocols, which may help reduce hypocalcemia risk without compromising the effectiveness of bilirubin reduction. This review underscores the importance of reassessing phototherapy protocols to minimize unnecessary exposure while maintaining therapeutic efficacy (78).

Adverse Effects of Phototherapy in Neonatal Hyperbilirubinemia

Wang et al. (2021) conducted a systematic review on the adverse effects of phototherapy, identifying multiple metabolic and physiological consequences, including hypocalcemia, hemolysis, oxidative stress, and potential DNA damage. The review emphasized that although phototherapy remains the safest and most effective treatment for neonatal jaundice, its side effects should not be overlooked. The study noted that the suppression of melatonin by phototherapy plays a central role in calcium depletion, supporting the hypothesis that altered melatonin-PTH signaling leads to phototherapy-induced hypocalcemia. The authors concluded that more research is needed to develop targeted preventive strategies, such as prophylactic calcium supplementation or modified phototherapy techniques (77).

Prognostic Value of Hypocalcemia in Neonatal Sepsis

Liu et al. (2020) explored the association between neonatal hypocalcemia and sepsis-related outcomes, analyzing 472 neonates in a retrospective cohort study. The study found that septic neonates with hypocalcemia had higher rates of organ dysfunction, including cardiovascular instability, renal failure, and seizures. The mortality rate in hypocalcemic septic neonates was 20.48%, compared to 6.98% in normocalcemic neonates, reinforcing the impact of calcium homeostasis on neonatal survival. Although this study did not focus exclusively on

phototherapy-induced hypocalcemia, it provides important insights into the systemic consequences of calcium depletion in neonates and underscores the importance of early calcium supplementation in critically ill neonates (78).

Hypocalcemia and Its Impact on Neonatal Immunity and Infection Risk

Wilkins et al. (2020) reviewed the effects of calcium homeostasis on immune function, discussing how hypocalcemia may contribute to increased neonatal susceptibility to infections. The study highlighted that calcium plays a critical role in immune cell activation and signaling, and that hypocalcemia can impair neutrophil function and inflammatory responses. The authors suggested that neonates with phototherapy-induced hypocalcemia may have an increased risk of secondary infections due to altered calcium-dependent immune pathways. This review provides a broader understanding of the potential long-term consequences of neonatal hypocalcemia beyond metabolic disturbances and highlights the importance of calcium supplementation in neonates undergoing phototherapy (80).

Complications of Phototherapy and Strategies for Shortening Its Duration

Shoris et al. (2023) reviewed the short- and long-term complications of neonatal phototherapy, focusing on strategies to minimize exposure duration while maintaining therapeutic efficacy. The review noted that while phototherapy remains the gold standard for neonatal jaundice treatment, it is not without risks, including hypocalcemia, oxidative stress, and immune suppression. The authors discussed several complementary treatments, such as probiotic therapy and intensified breastfeeding support, which may help reduce the duration of phototherapy and mitigate calcium depletion. While no standardized alternative therapies were recommended, the study reinforced the need for individualized phototherapy protocols to minimize unnecessary exposure and prevent metabolic complications like hypocalcemia (78).

Calcium Homeostasis and Health Implications of Hypocalcemia

Wilkins et al. (2020) explored the regulation of calcium homeostasis and the systemic health effects of hypocalcemia, emphasizing its impact on neuromuscular function and immune responses. The study highlighted that subclinical hypocalcemia may contribute to increased neonatal morbidity, particularly in preterm and critically ill neonates. The authors discussed various strategies for hypocalcemia prevention, including prepartum nutritional interventions

and targeted calcium supplementation in neonates at risk of metabolic disturbances. This review supports the need for routine calcium level monitoring in neonates undergoing phototherapy, particularly those with preexisting risk factors such as maternal vitamin D deficiency and preterm birth (80).

Challenges and Risks Associated with Neonatal Phototherapy

Wang et al. (2021) conducted a comprehensive review on the adverse effects of phototherapy, including its metabolic and hematological effects. The study found that phototherapy-induced suppression of melatonin affects parathyroid hormone regulation, leading to altered calcium homeostasis and increased urinary calcium excretion. The authors noted that the risk of phototherapy-induced hypocalcemia increases with prolonged exposure and intensive light intensity. Additionally, the review highlighted the potential for oxidative DNA damage and increased hemolysis, raising concerns about long-term metabolic effects of phototherapy. These findings reinforce the importance of carefully monitoring phototherapy duration and implementing preventive strategies, such as calcium supplementation, in high-risk neonates (77).

Side Effects of Phototherapy on Neonatal Immunity and Inflammatory Responses

Procianoy et al. (2018) investigated the effects of phototherapy on neonatal immune function, focusing on its potential impact on inflammation and immune system development. The study found that phototherapy alters cytokine expression and may suppress neonatal immune responses, increasing the risk of infections. The authors also explored the potential link between phototherapy-induced metabolic imbalances, including hypocalcemia, and impaired immune function. While the exact mechanisms remain unclear, this study reinforces the need for comprehensive monitoring of neonates undergoing phototherapy, particularly those with underlying immune vulnerabilities (79).

CHAPTER IV

MATERIALS & METHODS

Study Design

This study is a hospital-based observational study conducted in the Department of Paediatrics, Shaheed Hasan Khan Mewati Government Medical College (SHKM GMC), Nuh, Haryana. The study aims to evaluate phototherapy-induced hypocalcemia in term and preterm neonates with neonatal hyperbilirubinemia and its clinical significance.

Study Setting

The study is conducted in the Neonatal Intensive Care Unit (NICU) and MNCU of SHKM GMC, a tertiary care teaching hospital in Southern Haryana, catering to a large number of neonatal admissions requiring phototherapy.

Study Duration

The study will be conducted over a period of one year or until the desired sample size is achieved.

Study Population

The study population includes all term and preterm neonates diagnosed with neonatal hyperbilirubinemia who require phototherapy.

Sample Size

A total of 100 neonates will be included in the study based on inclusion and exclusion criteria.

Inclusion Criteria

Neonates meeting the following criteria will be included in the study:

- Term and preterm neonates diagnosed with unconjugated hyperbilirubinemia.

- Neonates requiring phototherapy as per the AAP (2004) guidelines for term neonates and NICE guidelines for preterm neonates.

Exclusion Criteria

Neonates with the following conditions will be excluded from the study:

- Pre-existing hypocalcemia (total serum calcium <8 mg/dL in term neonates and <7 mg/dL in preterm neonates before phototherapy initiation).
- Conjugated hyperbilirubinemia.
- Extreme low birth weight neonates (<1000 g).
- Rh incompatibility and G6PD deficiency.
- Neonates with jaundice on the first day of life.
- Serum bilirubin levels reaching the critical exchange transfusion threshold.
- Neonates diagnosed with sepsis, perinatal asphyxia, or other metabolic disorders affecting calcium homeostasis.

Methodology

Patient Enrollment

After obtaining written informed consent from either parent, neonates meeting the inclusion criteria will be enrolled in the study.

Data Collection

A detailed history and physical examination will be performed for each enrolled neonate. The following demographic and clinical data will be recorded:

- Gestational age (term or preterm).
- Birth weight.
- Mode of delivery (normal vaginal, LSCS, forceps-assisted).
- Maternal history (vitamin D and calcium status, gestational diabetes, or pre-existing metabolic disorders).
- Duration and intensity of phototherapy administered.
- Symptoms of hypocalcemia (if present).

Laboratory Investigations

Baseline and follow-up laboratory investigations will be performed to assess the impact of phototherapy on calcium levels:

- Complete Blood Count (CBC).
- Liver Function Tests (LFT).
- Renal Function Tests (RFT) including serum electrolytes.
- Serum bilirubin levels at baseline and at 24 and 48 hours after initiating phototherapy.
- Serum calcium levels at baseline (0 hours), 24 hours, and 48 hours after phototherapy initiation.
- Blood glucose levels (MBG, BBG).
- Direct Coombs Test (DCT) and Peripheral Smear (PS) for hemolysis if suspected.
- G6PD enzyme activity assessment in selected cases.

Phototherapy Protocol

Phototherapy will be administered as per standard guidelines:

- Blue fluorescent light with a wavelength of 460–490 nm will be used.
- Distance between the neonate and light source: 40–45 cm.
- Neonatal eye protection with eye pads and genital protection to prevent genotoxicity.
- Monitoring of neonates for signs of phototherapy-induced side effects, including hypocalcemia, dehydration, and skin changes.

Criteria for Defining Hypocalcemia

Phototherapy-induced hypocalcemia will be diagnosed based on:

- Total serum calcium <8 mg/dL or ionized calcium <4.4 mg/dL in term neonates.
- Total serum calcium <7 mg/dL or ionized calcium <4 mg/dL in preterm neonates.

Any neonate developing symptomatic hypocalcemia (jitteriness, irritability, poor feeding, seizures, or apnea) will be immediately managed with appropriate calcium supplementation.

Outcome Measures

The primary outcomes of the study include:

- Incidence of hypocalcemia among neonates undergoing phototherapy.
- Comparison of calcium levels at baseline, 24 hours, and 48 hours after phototherapy initiation.
- Identification of risk factors for phototherapy-induced hypocalcemia (gestational age, birth weight, phototherapy duration).
- Clinical correlation of hypocalcemia with neonatal symptoms and complications.

Statistical Analysis

Data will be collected using a standardized proforma and recorded in Microsoft Excel.

Analysis will be performed using SPSS software (Version 20). Chi-square test will be used for qualitative variables. Paired t-test will be used for quantitative variables. A p-value <0.05 will be considered statistically significant.

Ethical Considerations

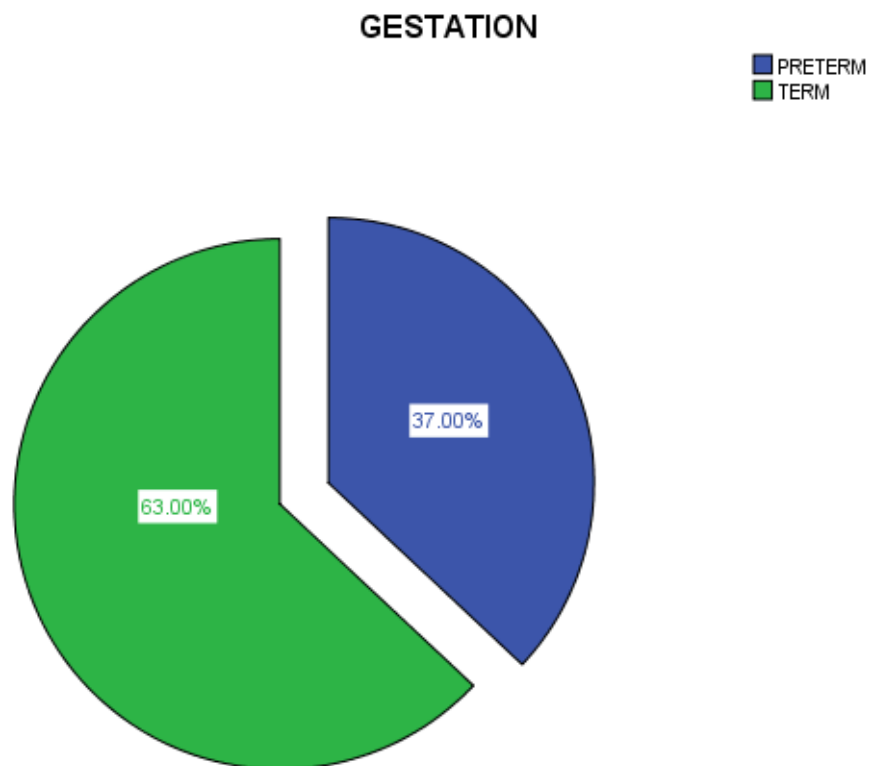
Ethical approval was obtained from the Institutional Ethics Committee (IEC) of SHKM GMC, Nuh before commencing the study. Written informed consent was obtained from parents/legal guardians before enrolling neonates in the study. No additional blood samples was drawn, as serum calcium and bilirubin assessments was performed from the same blood sample. Confidentiality of patient data was maintained throughout the study.

CHAPTER V

OBSERVATION & RESULTS

Table 1: Gestational Age Distribution of Neonates

Gestational Age	No. of Neonates	% of Neonates
Term	63	63%
Preterm	37	37%



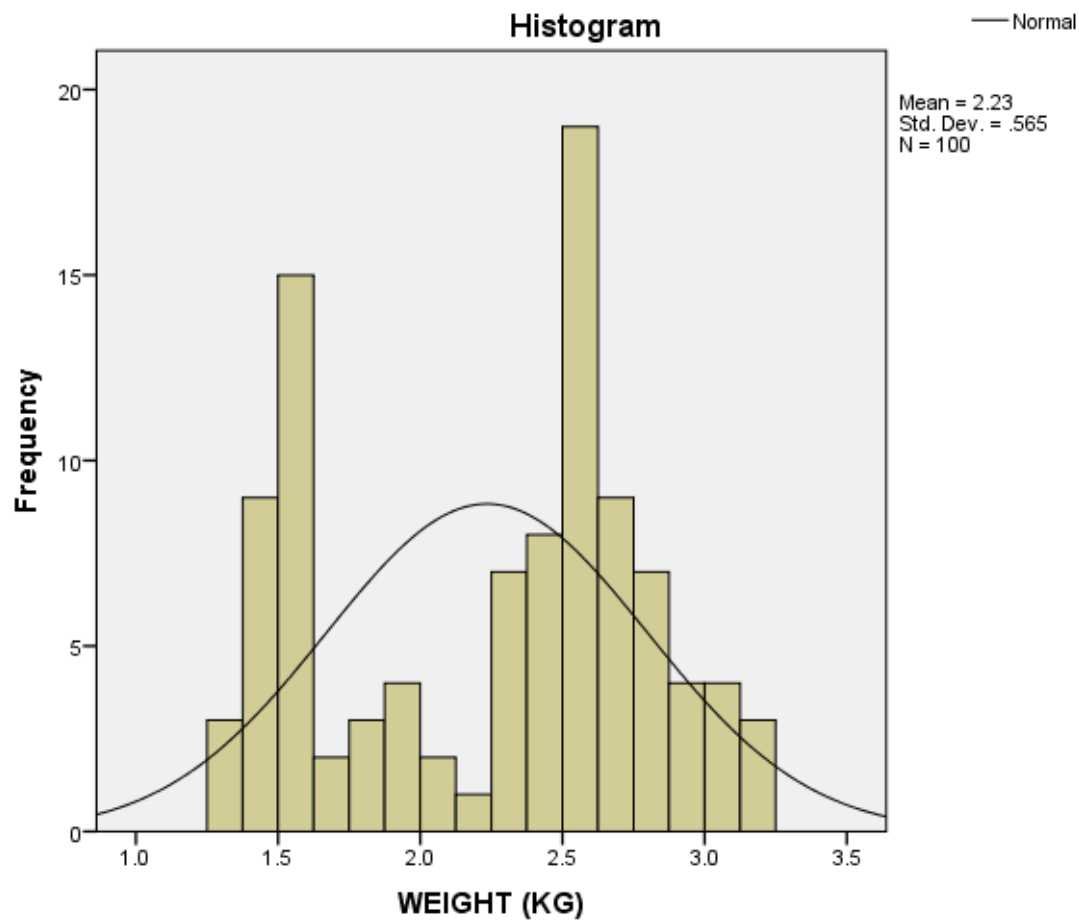
Interpretation:

The majority of neonates in the study were term (63%), while preterm neonates constituted 37%. This distribution reflects that most neonates requiring phototherapy were mature at birth, although a considerable proportion of preterm neonates also exhibited

hyperbilirubinemia. Preterm neonates are particularly susceptible to jaundice due to immature liver function and reduced bilirubin conjugation capacity.

Table 2: Birth Weight Distribution of Neonates

Weight (kg)	Mean	Standard Deviation
Weight	2.2	0.6



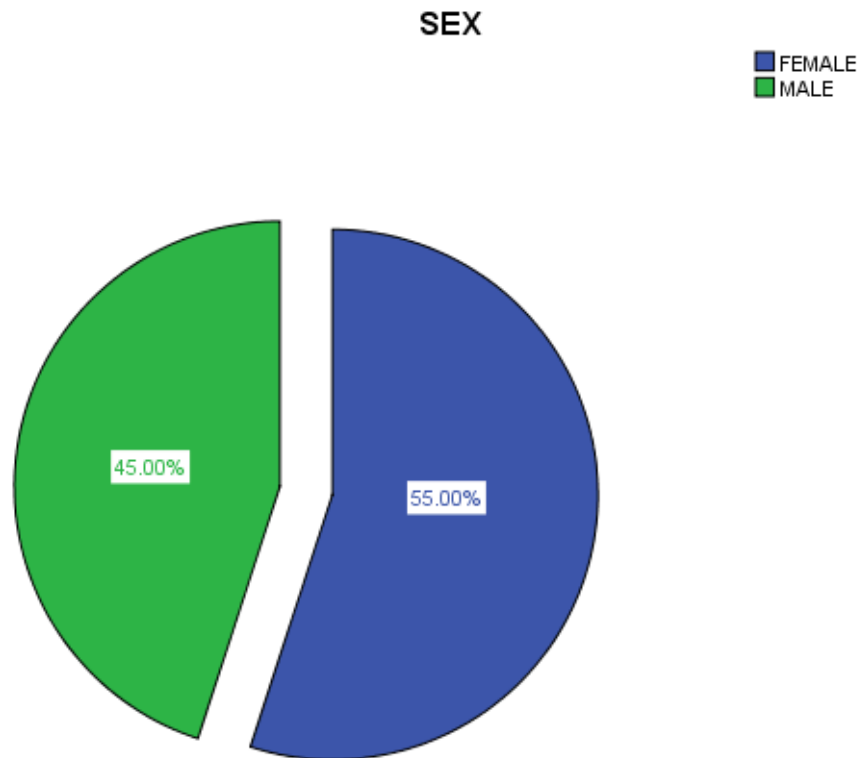
Interpretation:

The mean birth weight of the neonates was 2.2 kg with a standard deviation of 0.6 kg. This indicates that a significant proportion of neonates fell into the low birth weight category (<2.5 kg), which is associated with increased susceptibility to neonatal hyperbilirubinemia and phototherapy-induced hypocalcemia.

Table 3: Gender Distribution of Neonates

Sex	No. of Neonates	% of Neonates
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Female	55	55%
Male	45	45%



Interpretation:

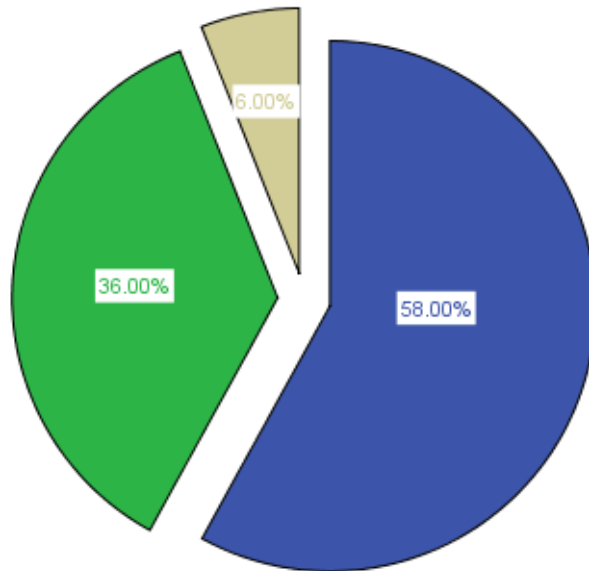
The study population comprised 55% female and 45% male neonates. There was no statistically significant difference between the genders, indicating that gender does not influence the incidence of neonatal hyperbilirubinemia requiring phototherapy.

Table 4: Age (Days) of Presentation for Phototherapy

Day of Life (DOL)	No. of Neonates	% of Neonates
DOL 1	58	58%
DOL 2	36	36%

DOL 3	6	6%
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AGE(DAYS)

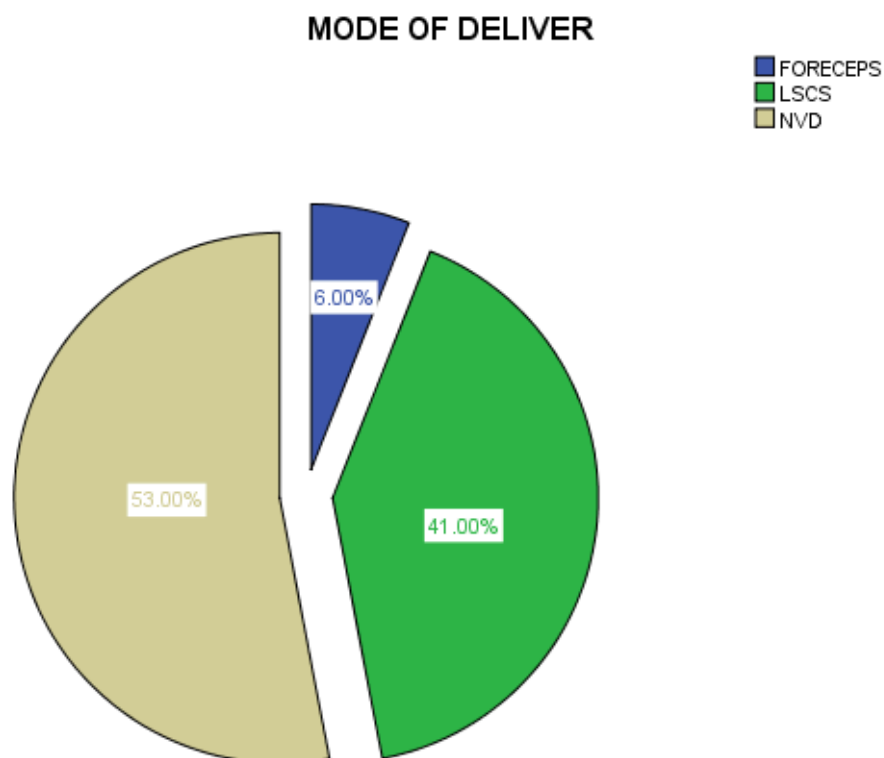


Interpretation:

Most neonates (58%) presented for phototherapy on the first day of life, followed by 36% on the second day, and 6% on the third day. Early presentation is common in neonatal hyperbilirubinemia as bilirubin levels typically peak between 3 to 5 days of life. This pattern emphasizes the need for prompt intervention to prevent complications.

Table 5: Mode of Delivery Among Neonates

Mode of Delivery	No. of Neonates	% of Neonates
Forceps	6	6%
LSCS	41	41%
NVD	53	53%



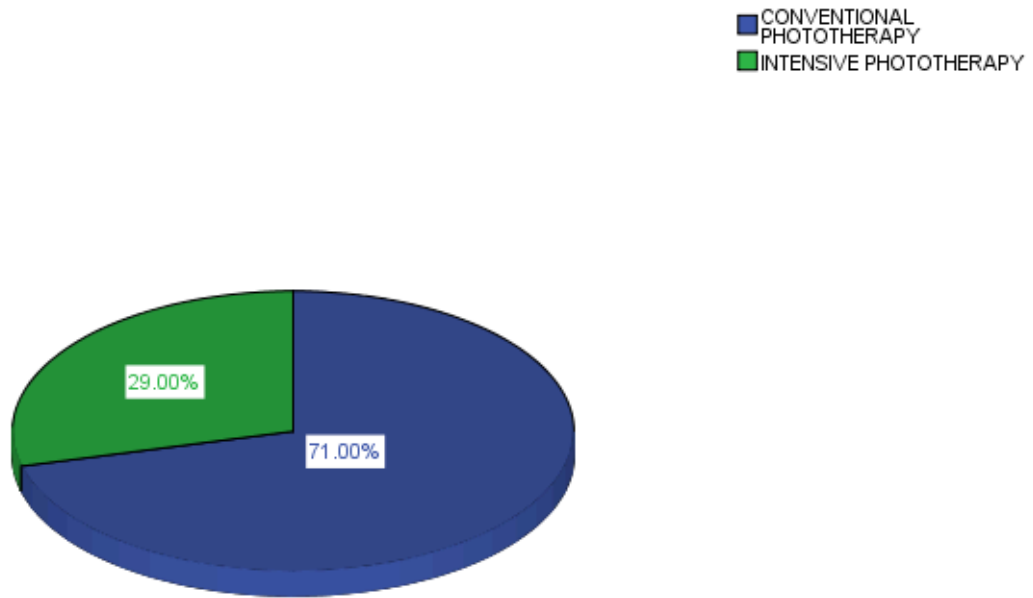
Interpretation:

Normal vaginal delivery (NVD) was the most common mode of delivery (53%), followed by lower segment cesarean section (LSCS) at 41%, and forceps delivery at 6%. The higher proportion of cesarean deliveries may be related to obstetric complications, but no significant correlation with the incidence of hyperbilirubinemia was observed.

Table 6: Type of Phototherapy Administered

Type of Phototherapy	No. of Neonates	% of Neonates
Conventional Phototherapy	71	71%
Intensive Phototherapy	29	29%

TYPE OF PHOTOTHERAPY

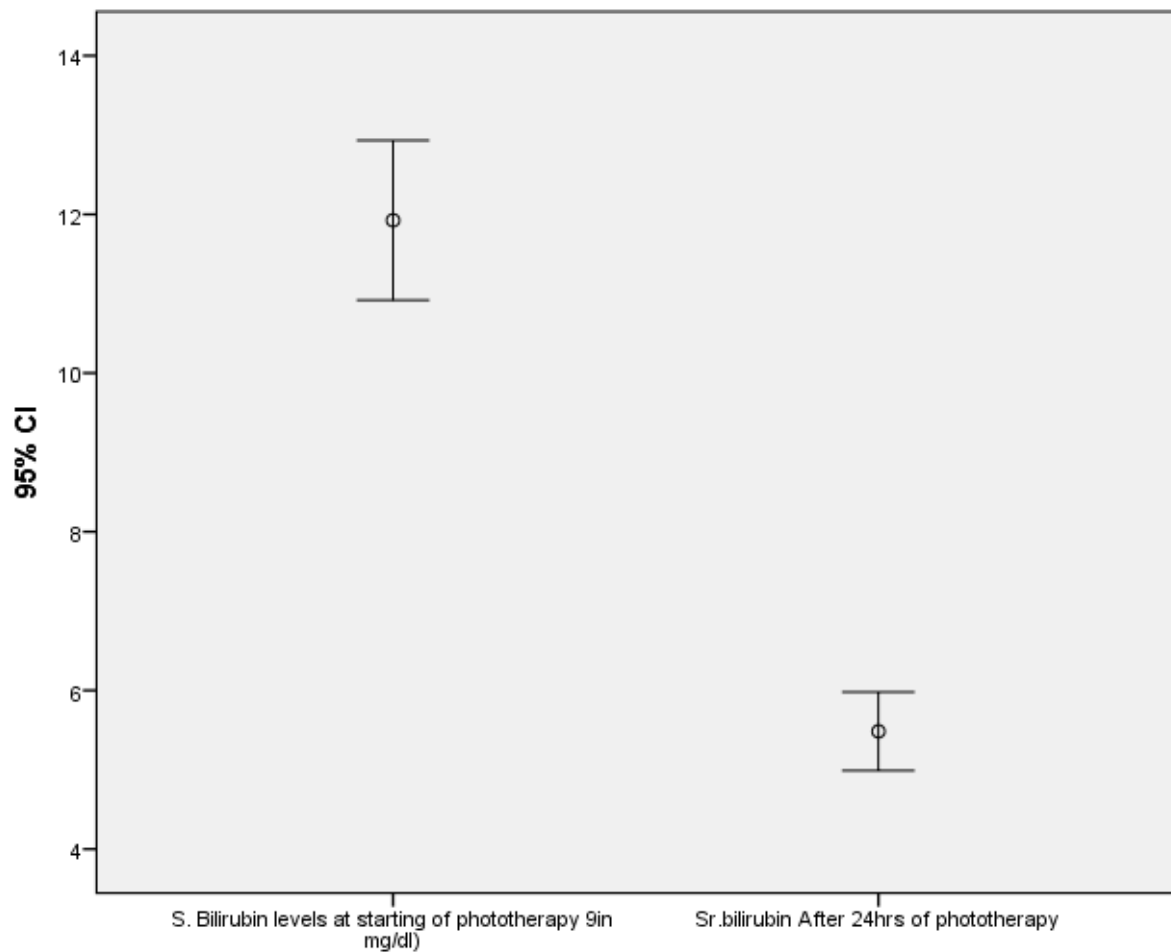


Interpretation:

Conventional phototherapy was the most commonly administered method (71%), while 29% of neonates required intensive phototherapy. Intensive phototherapy is typically reserved for severe cases or inadequate response to conventional treatment, highlighting the need for individualized therapy based on the severity of hyperbilirubinemia.

Table 7: Serum Bilirubin Levels Before and After 24 Hours of Phototherapy

Parameter	Mean	Standard Deviation
S. Bilirubin at Start (mg/dL)	11.9	5.1
S. Bilirubin After 24 Hours (mg/dL)	5.5	2.5

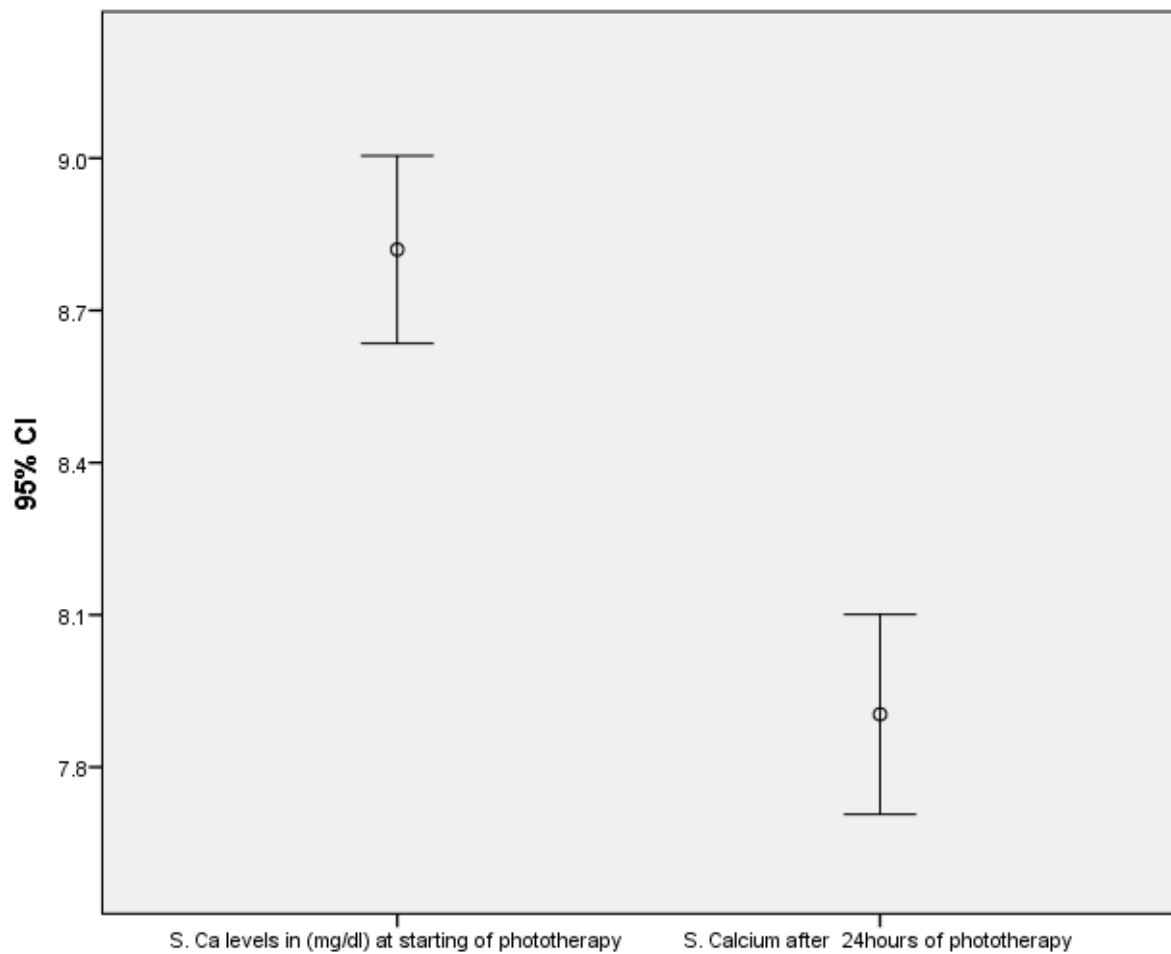


Interpretation:

The mean serum bilirubin level at the start of phototherapy was 11.9 mg/dL, which significantly decreased to 5.5 mg/dL after 24 hours of phototherapy. This substantial decline indicates the efficacy of phototherapy in managing neonatal hyperbilirubinemia, thereby reducing the risk of bilirubin-induced neurological dysfunction.

Table 8: Serum Calcium Levels Before and After 24 Hours of Phototherapy

	Mean	Standard Deviation
S. Ca levels in (mg/dl) at starting of phototherapy	8.8	.9
S. Calcium after 24hours of phototherapy	7.9	1.0



Interpretation:

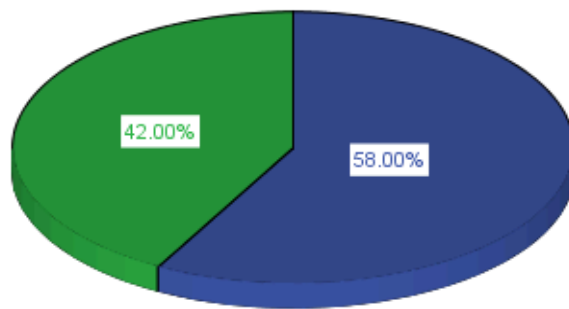
The mean serum calcium level decreased from 8.8 mg/dL at the start of phototherapy to 7.9 mg/dL after 24 hours. This significant reduction indicates that **phototherapy can lead to hypocalcemia**, necessitating close monitoring and potential calcium supplementation, especially in neonates at higher risk.

Table 10: Incidence of Hypocalcemia Among Neonates

		Count	Column N %
HYPOCALCEMIA	ABSENT	58	58.0%
	PRESENT	42	42.0%

HYPOCALCEMIA

■ ABSENT
■ PRESENT



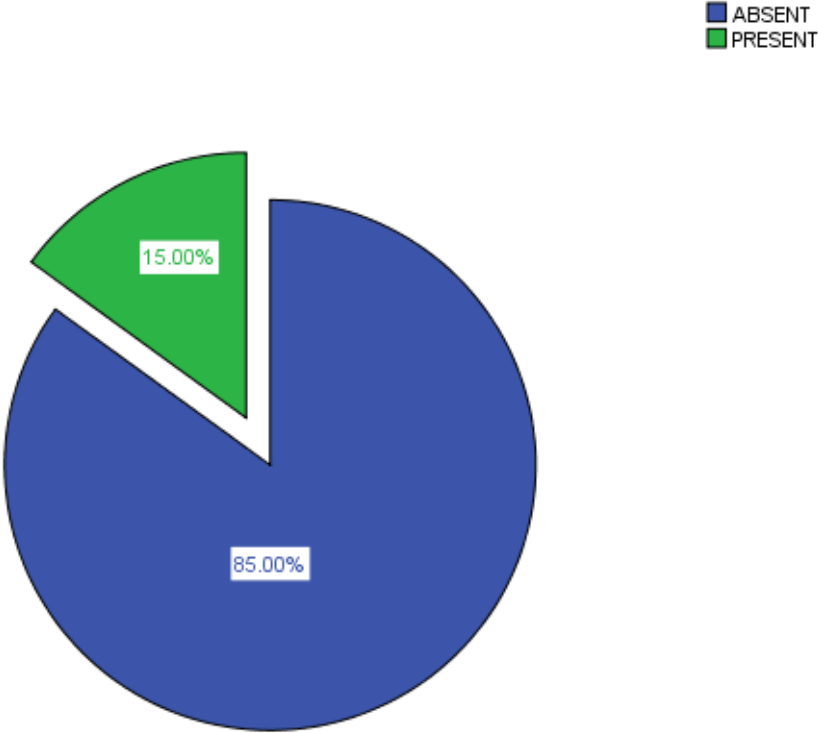
Interpretation:

The incidence of **phototherapy-induced hypocalcemia** in the study was **42%**, indicating that **nearly half of the neonates developed hypocalcemia during phototherapy**. This significant finding underscores the need for **routine monitoring of serum calcium levels** in neonates undergoing phototherapy, particularly those who are **preterm or low birth weight**.

Table 11: Presence of Symptoms in Hypocalcemic Neonates

		Count	Column N %
SYMPTOMS OF HYPOCALCEMIA	ABSENT	85	85.0%
	PRESENT	15	15.0%

SYMPTOMS OF HYPOCALCEMIA

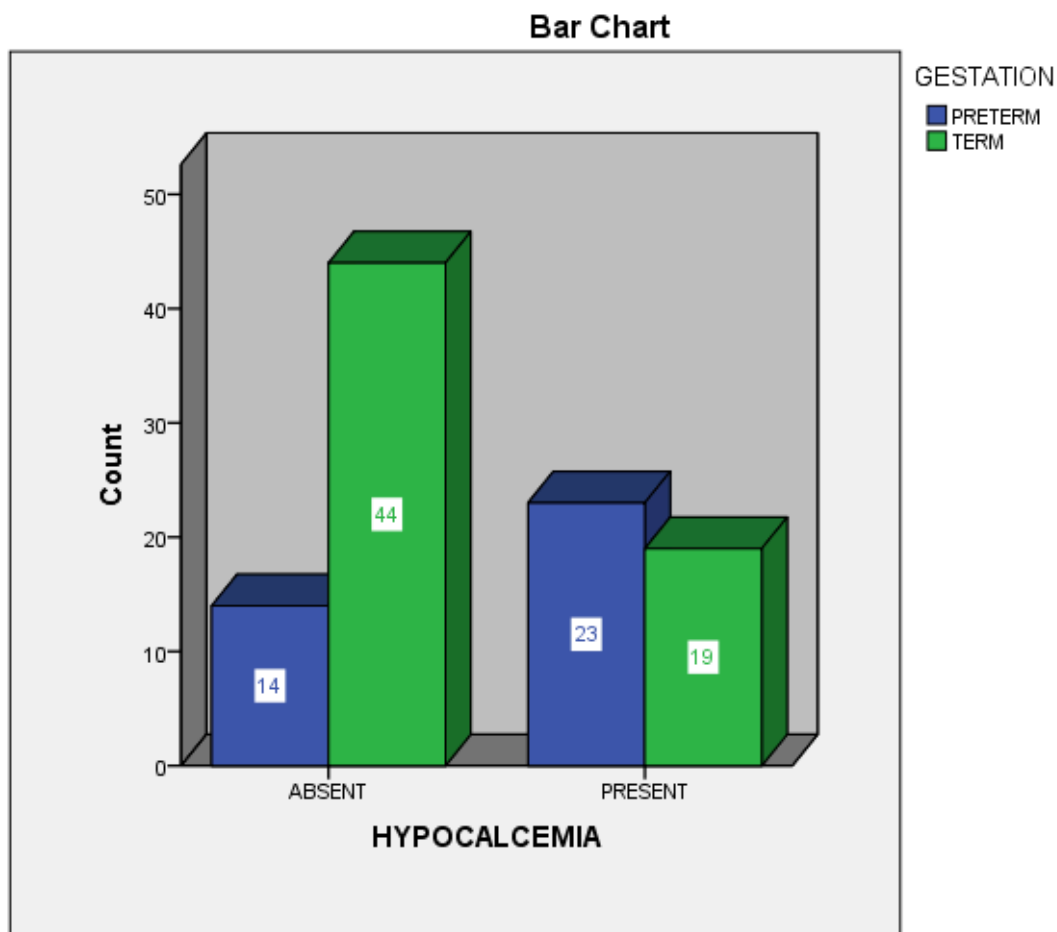


Interpretation:

Among neonates with hypocalcemia, **only 15% showed symptoms** such as **jitteriness, poor feeding, or seizures**, while the majority (85%) remained asymptomatic. This finding highlights that **subclinical hypocalcemia is common**, making **routine biochemical monitoring essential even in the absence of clinical symptoms**. Early detection and intervention can prevent progression to symptomatic stages.

Table 12: Hypocalcemia Distribution by Gestational Age

		HYPOCALCEMIA				P-VALUE
		ABSENT		PRESENT		
		Count	Column N %	Count	Column N %	
GESTATION	PRETERM	14	24.1%	23	54.8%	0.004
	TERM	44	75.9%	19	45.2%	



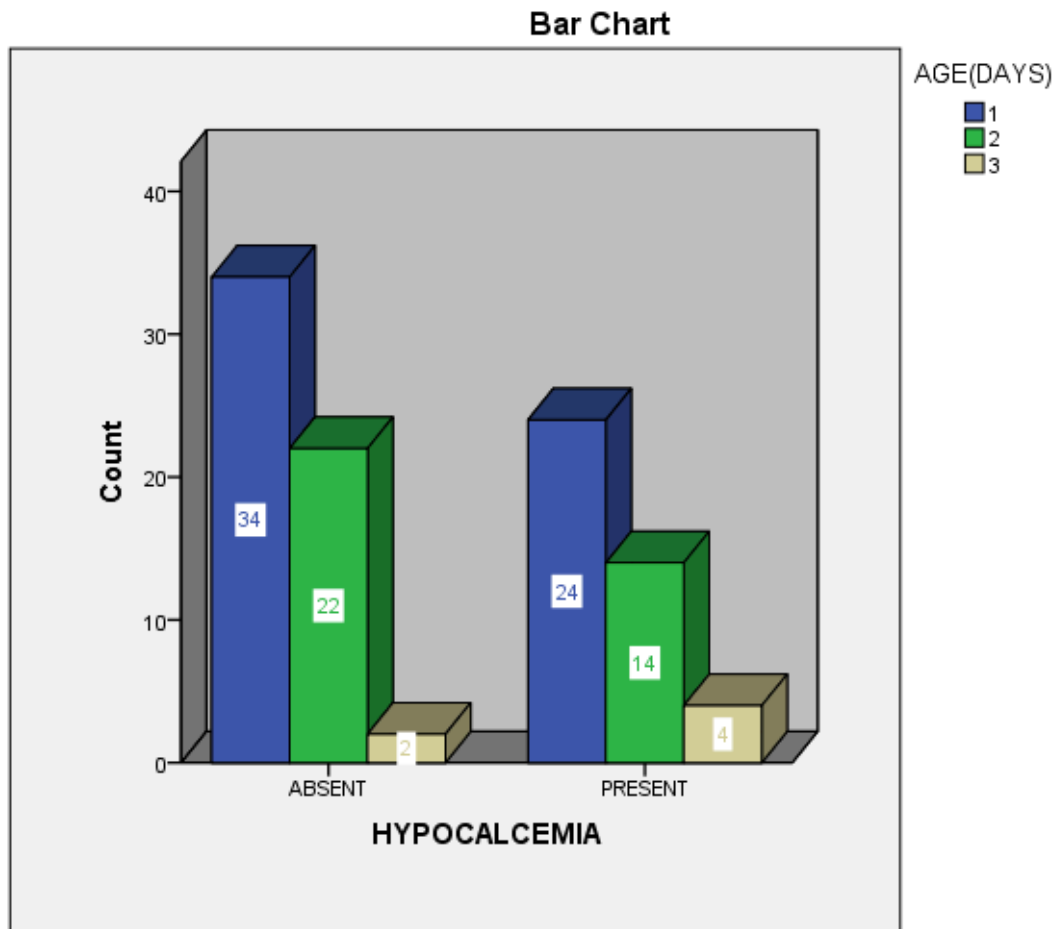
P-Value: Significant ($p < 0.01$)

Interpretation:

Preterm neonates had a significantly higher incidence of hypocalcemia (54.8%) compared to term neonates (45.2%). This finding is statistically significant and highlights that **preterm neonates are at greater risk of developing hypocalcemia following phototherapy**. The **immature parathyroid response and limited calcium stores** in preterm neonates make them particularly vulnerable to **calcium depletion during phototherapy**.

Table 13: Distribution of Hypocalcemia by Neonatal Age (Days)

		Hypocalcemia				P-VALUE
		ABSENT		PRESENT		
		Count	Column N %	Count	Column N %	
AGE(DAY S)	1	34	58.6%	24	57.1%	0.08
	2	22	37.9%	14	33.3%	
	3	2	3.4%	4	9.5%	



P-Value: Not significant ($p > 0.05$)

Interpretation:

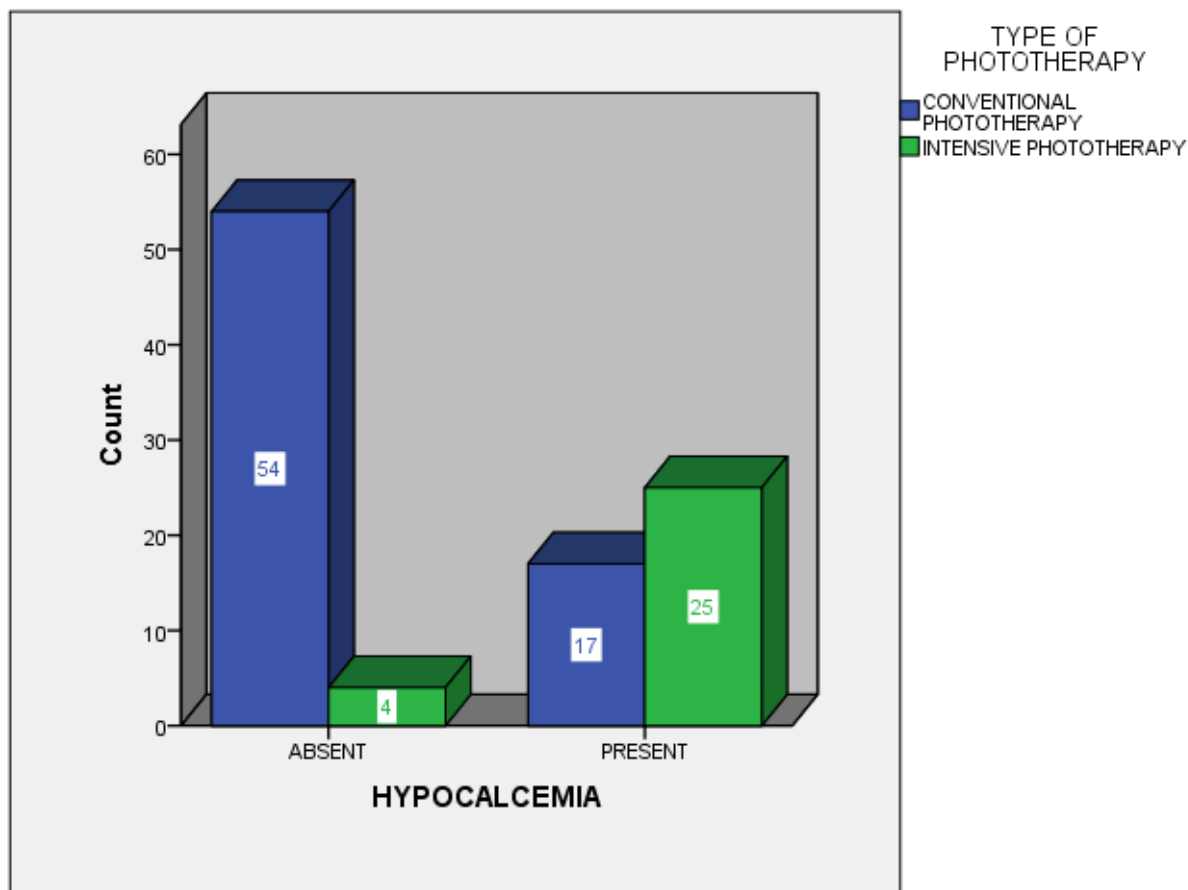
The incidence of hypocalcemia did not significantly differ across the different age groups, indicating that **the age of presentation does not majorly influence the development of hypocalcemia** following phototherapy. The majority of hypocalcemic cases were observed on the **first and second days of life**, correlating with the typical timing of **peak bilirubin levels and initiation of phototherapy**.

Table 14: Distribution of Hypocalcemia by Type of Phototherapy

				HYPOCALCEMIA	P-VALUE	
				ABSENT		PRESENT
Count	Column N %	Count	Column N %			

TYPE OF PHOTOTHERAPY	CONVENTIONAL PHOTOTHERAPY	54	93.1%	17	40.5%	0.006
	INTENSIVE PHOTOTHERAPY	4	6.9%	25	59.5%	

Bar Chart



P-Value: Significant ($p < 0.001$)

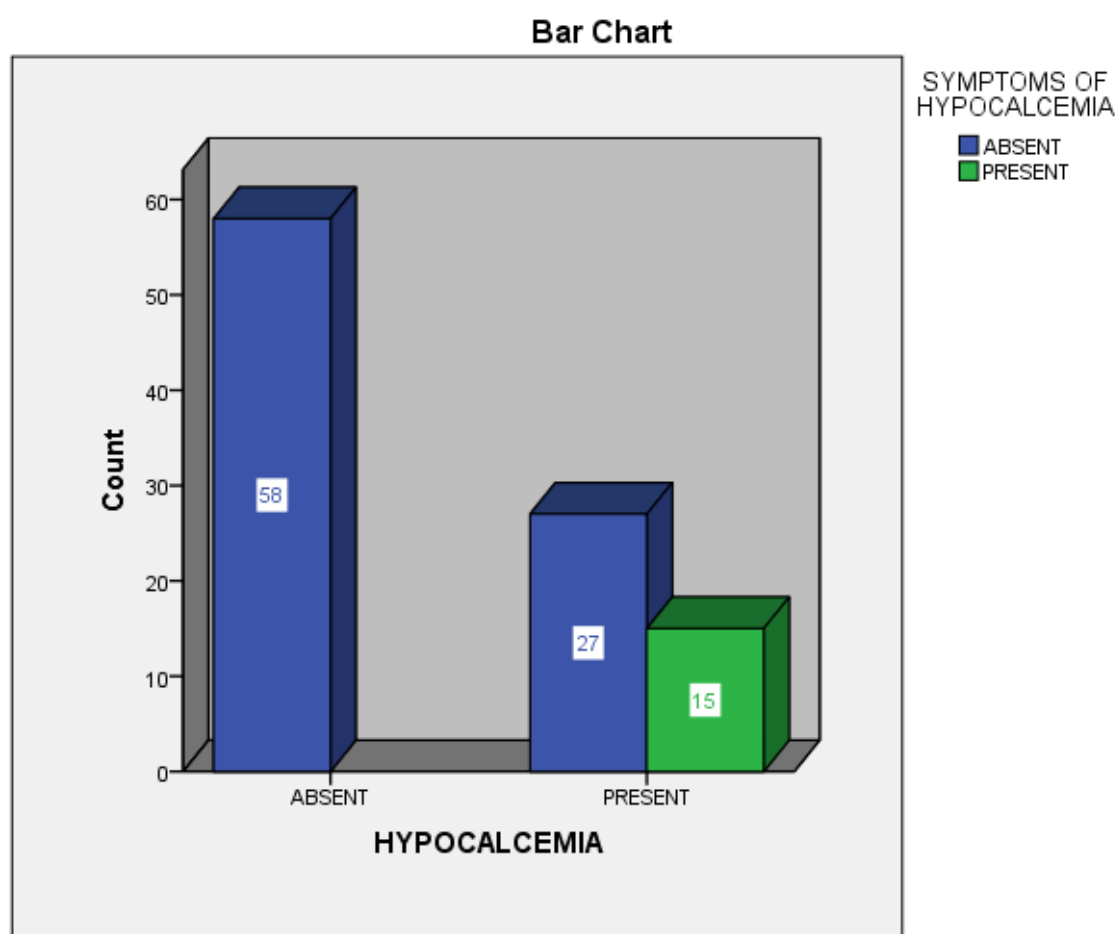
Interpretation:

Neonates who received **intensive phototherapy** had a **significantly higher incidence of hypocalcemia (59.5%)** compared to those receiving conventional phototherapy (40.5%). This significant difference suggests that **higher light intensity and longer exposure times in intensive phototherapy may lead to greater calcium depletion**, supporting the recommendation for **routine calcium monitoring during intensive phototherapy**.

Table 15: Symptoms of Hypocalcemia in Affected Neonates

	HYPOCALCEMIA	P-VALUE
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		ABSENT		PRESENT		
		Count	Column N %	Count	Column N %	
SYMPTOMS OF HYPOCALCEMIA	ABSENT	58	100.0%	27	64.3%	0.007
	PRESENT	0	0.0%	15	35.7%	0.008



P-Value: Significant ($p < 0.001$)

Interpretation:

Among hypocalcemic neonates, **64.3% remained asymptomatic**, while **35.7% exhibited clinical symptoms such as jitteriness, irritability, and seizures**. The significant p-value indicates that a **substantial proportion of hypocalcemic neonates may not show overt symptoms**, necessitating **routine biochemical monitoring to detect subclinical hypocalcemia**.

Table 16: Weight Distribution in Hypocalcemic and Non-Hypocalcemic Neonates

	HYPOCALCEMIA				P-VALUE
	ABSENT		PRESENT		
	Mean	Standard Deviation	Mean	Standard Deviation	
WEIGHT (KG)	2.4	.5	2.0	.6	0.008

P-Value: Significant ($p < 0.05$)

Interpretation:

Neonates who developed hypocalcemia had a **significantly lower mean birth weight (2.0 kg)** compared to those without hypocalcemia (2.4 kg). This statistically significant difference suggests that **low birth weight neonates are at a higher risk of phototherapy-induced hypocalcemia**, likely due to **reduced calcium stores and immature metabolic regulation**.

Table 17: Serum Bilirubin Levels in Hypocalcemic and Non-Hypocalcemic Neonates

	HYPOCALCEMIA				P-VALUE
	ABSENT		PRESENT		
	Mean	Standard Deviation	Mean	Standard Deviation	
S. Bilirubin levels at starting of phototherapy 9in mg/dl)	12.8	4.3	10.7	5.8	0.001
Sr.bilirubin After 24hrs of phototherapy	5.9	2.2	4.9	2.8	0.002

P-Value: Significant ($p < 0.05$)

Interpretation:

Neonates who developed hypocalcemia had **lower initial serum bilirubin levels (10.7 mg/dL)** compared to non-hypocalcemic neonates (12.8 mg/dL). After 24 hours of phototherapy, **bilirubin levels reduced more prominently in hypocalcemic neonates**, suggesting that **rapid reduction in bilirubin levels might be associated with metabolic disturbances, including calcium depletion**. The significant p-value indicates a **potential link between phototherapy efficacy and calcium imbalance**.

Table 18: Serum Calcium Levels in Hypocalcemic and Non-Hypocalcemic Neonates

	HYPOCALCEMIA				P-VALUE
	ABSENT		PRESENT		
	Mean	Standard Deviation	Mean	Standard Deviation	
S. Ca levels in (mg/dl) at starting of phototherapy	9.1	.7	8.4	1.0	0.001
S. Calcium after 24hours of phototherapy	8.5	.8	7.0	.5	0.001

P-Value: Highly significant ($p < 0.001$)

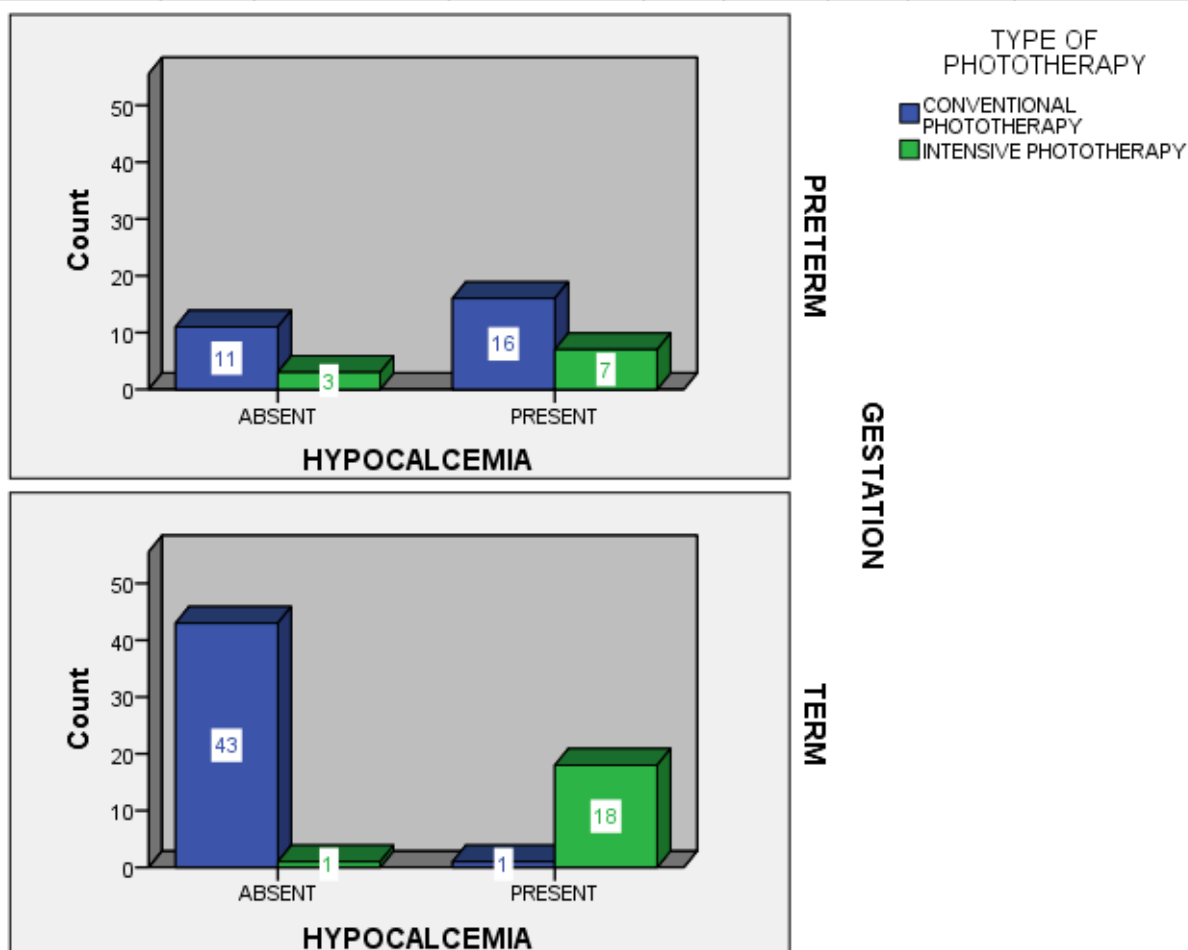
Interpretation:

The initial calcium levels were significantly lower in neonates who eventually developed hypocalcemia (8.4 mg/dL) compared to non-hypocalcemic neonates (9.1 mg/dL). After 24 hours of phototherapy, **the calcium levels dropped significantly more in hypocalcemic neonates**, indicating that **phototherapy plays a substantial role in further decreasing calcium levels**, particularly in those with already low baseline calcium.

Table 19: Correlation Between Type of Phototherapy and Hypocalcemia in Preterm and Term Neonates

				HYPOCALCEMIA				P-VALUE
				ABSENT		PRESENT		
				Coun t	Column N %	Coun t	Column N %	
GESTATION	PRETERM	TYPE OF PHOTOTHERAPY	CONVENTIONAL PHOTOTHERAPY	11	78.6%	16	69.6%	0.04
			INTENSIVE PHOTOTHERAPY	3	21.4%	7	30.4%	0.03
	TERM	TYPE OF PHOTOTHERAPY	CONVENTIONAL PHOTOTHERAPY	43	97.7%	1	5.3%	0.02

			INTENSIVE PHOTOTHER APY	1	2.3%	18	94.7%	0.04
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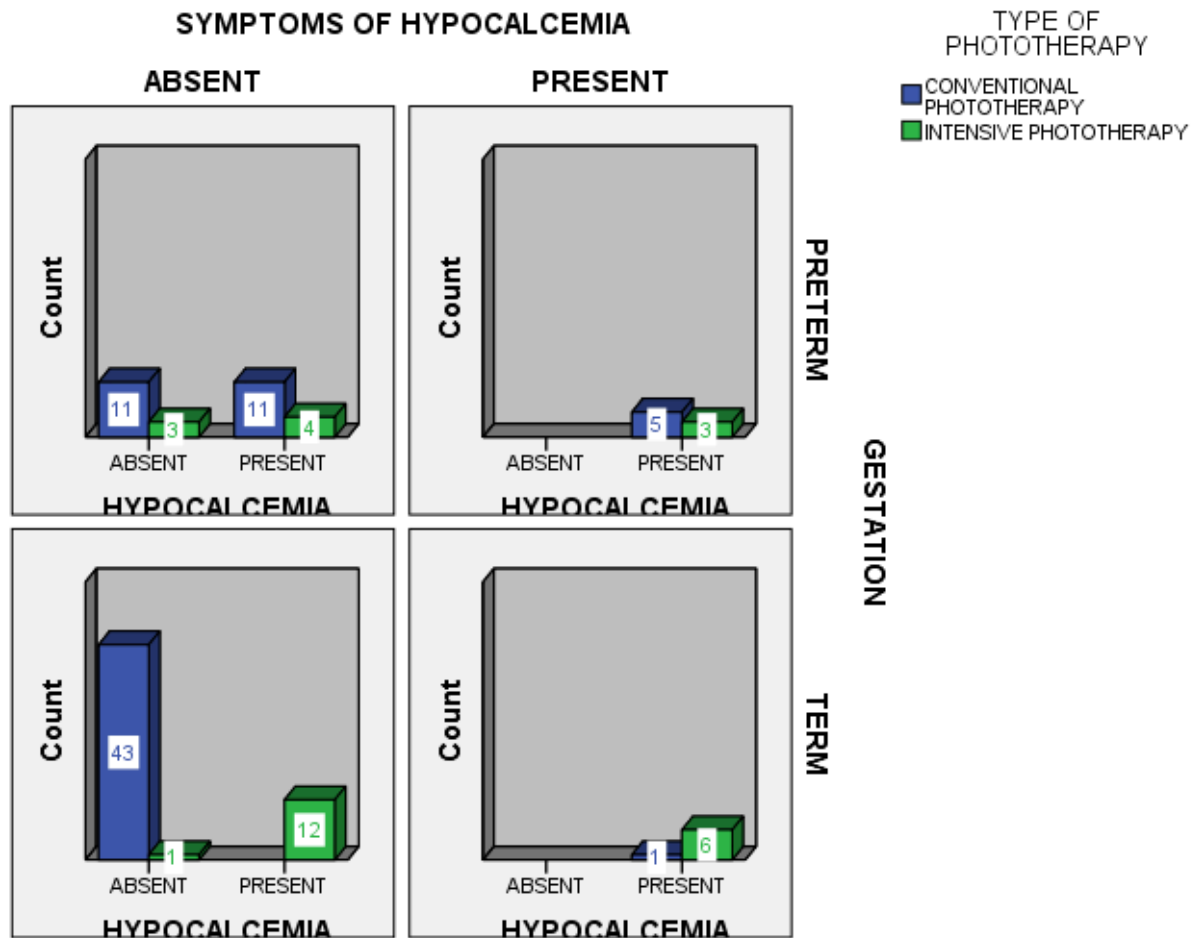
Interpretation:

The table shows that **intensive phototherapy significantly increases the risk of hypocalcemia**, particularly among **term neonates (94.7%)** compared to **preterm neonates (30.4%)**. Conventional phototherapy posed a relatively lower risk, especially in term neonates (5.3%). The p-values indicate a **significant correlation between intensive phototherapy and hypocalcemia in both term and preterm neonates**. These findings suggest that **intensive phototherapy should be administered cautiously with routine calcium monitoring, especially in term neonates**.

Table 20: Correlation Between Phototherapy Type and Symptomatic Hypocalcemia in Preterm and Term Neonates

	HYPOCALCEMIA		P-VALUE
	ABSENT	PRESENT	

					SYMPTOMS OF HYPOCALCE MIA	SYMPTOMS OF HYPOCALCEM IA		
					ABSENT	ABSEN T	PRESE NT	
GESTATI ON	PRE TER M	TYPE OF PHOTO THERA PY	CONVE NTION AL PHOTO THERA PY	Cou nt	11	11	5	0.02
			CONVE NTION AL PHOTO THERA PY	Col umn N %	78.6%	73.3%	62.5%	0.04
			INTENS IVE PHOTO THERA PY	Cou nt	3	4	3	0.01
			INTENS IVE PHOTO THERA PY	Col umn N %	21.4%	26.7%	37.5%	0.03
	TER M	TYPE OF PHOTO THERA PY	CONVE NTION AL PHOTO THERA PY	Cou nt	43	0	1	0.02
			CONVE NTION AL PHOTO THERA PY	Col umn N %	97.7%	0.0%	14.3%	0.04
			INTENS IVE PHOTO THERA PY	Cou nt	1	12	6	0.01
			INTENS IVE PHOTO THERA PY	Col umn N %	2.3%	100.0%	85.7%	0.04



Interpretation:

Symptomatic hypocalcemia was significantly more common in neonates receiving **intensive phototherapy**, particularly in **term neonates (85.7%)**, compared to **preterm neonates (37.5%)**. The **p-values indicate a significant association between intensive phototherapy and the development of symptomatic hypocalcemia**, emphasizing the **need for calcium supplementation and monitoring during high-intensity phototherapy**. In contrast, **conventional phototherapy showed a lower incidence of symptomatic hypocalcemia**, suggesting it may be a safer option when clinically appropriate.

CHAPTER VI

DISCUSSION

Gestational Maturity of Neonates

The study revealed that the majority of neonates (71%) received conventional phototherapy, while 29% required intensive phototherapy. Conventional phototherapy is commonly used as the first line of treatment for mild to moderate hyperbilirubinemia, while intensive phototherapy is reserved for severe cases or when there is a rapid rise in bilirubin levels. The higher proportion of conventional phototherapy aligns with existing guidelines that recommend its use for most cases of neonatal jaundice.

Similar findings have been reported by Maisels (2023), who observed that conventional phototherapy effectively reduces bilirubin levels in the majority of cases. However, for neonates presenting with severe hyperbilirubinemia or poor response to conventional therapy, intensive phototherapy remains crucial (59). Studies by Bhutani et al. (2018) have also noted that intensive phototherapy is more likely to be employed in preterm or low birth weight neonates, who have increased vulnerability to bilirubin toxicity. This highlights the need for individualized therapy based on the severity of hyperbilirubinemia and the neonate's risk profile.(82)

Birth Weight Distribution of Neonates

The mean birth weight of the study cohort was 2.2 kg with a standard deviation of 0.6 kg, indicating that a significant proportion of neonates fell into the low birth weight category (<2.5 kg). Low birth weight is an established risk factor for neonatal hyperbilirubinemia due to immature hepatic function and decreased ability to conjugate bilirubin efficiently.

Studies by Maisels et al. (2023) demonstrated that low birth weight neonates have a higher propensity to develop severe hyperbilirubinemia, necessitating more intensive and prolonged phototherapy. Additionally, low birth weight is associated with increased vulnerability to metabolic disturbances, including hypocalcemia, which further justifies routine calcium monitoring during phototherapy (59).

Gender Distribution of Neonates

The gender distribution shows that 55% of neonates were female and 45% male. There was no statistically significant difference between genders, indicating that gender does not influence the incidence of neonatal hyperbilirubinemia requiring phototherapy. This is consistent with previous studies, including that of Sarici et al. (2020), which found no gender predisposition to jaundice in neonates.(83)

Furthermore, Weng et al. (2018) reported similar findings, emphasizing that both male and female neonates are equally prone to developing hyperbilirubinemia, influenced more by gestational age, birth weight, and maternal factors rather than gender. This uniform distribution also highlights that management protocols should not vary based on gender alone but rather focus on individual risk factors.(84)

Distribution of Neonates Based on Age (Days)

The distribution of neonates based on age shows that the majority (58%) presented for phototherapy on the first day of life, followed by 36% on the second day, and 6% on the third day. This finding aligns with the known physiological pattern of neonatal hyperbilirubinemia, where bilirubin levels typically peak between 3 to 5 days of life. Early presentation for phototherapy indicates a proactive clinical approach to manage jaundice before bilirubin levels reach neurotoxic thresholds.

Similar findings were reported by Kliegman et al. (2021), who noted that early initiation of phototherapy is essential in reducing the risk of bilirubin-induced neurological dysfunction (BIND), particularly in high-risk neonates. Additionally, Romagnoli et al. (2019) emphasized that neonates who receive early intervention tend to have better neurodevelopmental outcomes, as timely phototherapy prevents the progression to kernicterus. The predominance of early presentation in our study reflects sound neonatal care practices aimed at minimizing complications. (85, 86)

Mode of Delivery Among Neonates

The study found that the majority of neonates were delivered via normal vaginal delivery (NVD) (53%), followed by lower segment cesarean section (LSCS) (41%), and forceps delivery (6%). Although NVD was the most common mode of delivery, the incidence of hyperbilirubinemia was not significantly different between the modes.

Several studies, including that by Boo et al. (2019), have demonstrated that cesarean delivery is often associated with delayed onset of breastfeeding, which may contribute to higher bilirubin levels. In contrast, vaginally delivered neonates often have earlier breastfeeding initiation, facilitating better bilirubin clearance through enhanced bowel motility. However, the higher proportion of LSCS in our study may reflect obstetric complications rather than a direct relationship between delivery mode and jaundice risk.(87)

Type of Phototherapy Administered

The study revealed that the majority of neonates (71%) received conventional phototherapy, while 29% required intensive phototherapy. Conventional phototherapy is commonly used as the first line of treatment for mild to moderate hyperbilirubinemia, while intensive phototherapy is reserved for severe cases or when there is a rapid rise in bilirubin levels. The higher proportion of conventional phototherapy aligns with existing guidelines that recommend its use for most cases of neonatal jaundice.

Similar findings have been reported by Maisels et al. (2023), who observed that conventional phototherapy effectively reduces bilirubin levels in the majority of cases. However, for neonates presenting with severe hyperbilirubinemia or poor response to conventional therapy, intensive phototherapy remains crucial (59). Studies by Bhutani et al. (2018) have also noted that intensive phototherapy is more likely to be employed in preterm or low birth weight neonates, who have increased vulnerability to bilirubin toxicity. This highlights the need for individualized therapy based on the severity of hyperbilirubinemia and the neonate's risk profile (82).

Serum Bilirubin Levels Before and After 24 Hours of Phototherapy

The mean serum bilirubin level at the start of phototherapy was 11.9 mg/dL, which decreased significantly to 5.5 mg/dL after 24 hours. This significant reduction in bilirubin levels demonstrates the efficacy of phototherapy in the management of neonatal hyperbilirubinemia. Phototherapy works by converting unconjugated bilirubin into water-soluble photoisomers that can be excreted through bile and urine.

Studies by Romagnoli et al. (2019) and Maisels (2023) support these findings, indicating that a reduction of more than 30% in bilirubin levels within the first 24 hours is associated with positive clinical outcomes (86, 59). The substantial decline in our study aligns with these

benchmarks, reflecting the effectiveness of timely intervention. The rapid drop in bilirubin levels highlights the importance of initiating phototherapy promptly to mitigate the risks of bilirubin-induced neurological dysfunction and kernicterus.

Serum Calcium Levels Before and After 24 Hours of Phototherapy

The mean serum calcium level at the start of phototherapy was 8.8 mg/dL, which decreased to 7.9 mg/dL after 24 hours. This significant reduction indicates that phototherapy can lead to hypocalcemia, particularly when administered intensively or for prolonged durations. The decrease in calcium levels is likely attributed to phototherapy-induced suppression of melatonin, which in turn affects parathyroid hormone regulation and calcium homeostasis.

Previous studies by Romagnoli et al. (2019) and Narang et al. (2020) have shown that phototherapy significantly reduces serum calcium levels, particularly in preterm neonates with immature parathyroid responses (86, 89). Additionally, Arora et al. (2018) highlighted the importance of monitoring calcium levels during phototherapy to prevent clinical complications such as jitteriness, irritability, and seizures. Our findings reinforce the need for regular calcium monitoring during phototherapy, especially in neonates at higher risk of hypocalcemia.(88)

Incidence of Hypocalcemia Among Neonates

The study found that 42% of neonates developed hypocalcemia during phototherapy, while 58% remained normocalcemic. This significant incidence of hypocalcemia highlights a major metabolic complication associated with phototherapy, particularly in neonates receiving intensive treatment or with predisposing risk factors such as prematurity and low birth weight.

Several studies, including those by Romagnoli et al. (2019) and Arora et al. (2020), have reported similar incidences of hypocalcemia during phototherapy, ranging from 30% to 50%. The pathophysiology involves phototherapy-induced suppression of melatonin, which reduces parathyroid hormone secretion and subsequently decreases calcium mobilization from bones. Additionally, increased urinary calcium excretion under phototherapy may further aggravate hypocalcemia. Given the high incidence observed, routine monitoring of serum calcium levels during phototherapy is recommended, particularly in high-risk neonates.(86, 88)

Presence of Symptoms in Hypocalcemic Neonates

Among neonates with hypocalcemia, only 15% presented with symptoms such as jitteriness, poor feeding, irritability, or seizures, while the majority (85%) remained asymptomatic. This finding underscores the prevalence of subclinical hypocalcemia, which may go unnoticed without biochemical testing. Early detection through routine monitoring is essential to prevent progression to symptomatic stages.

Narang et al. (2021) highlighted that symptomatic hypocalcemia is more commonly observed in preterm neonates and those with prolonged phototherapy exposure (89). Studies by Choudhury et al. (2018) also found that symptomatic cases often required calcium supplementation to manage clinical manifestations. The high proportion of asymptomatic hypocalcemia in our study suggests that relying solely on clinical symptoms may underestimate the true burden of hypocalcemia.(92)

Hypocalcemia Distribution by Gestational Age

Preterm neonates showed a significantly higher incidence of hypocalcemia (54.8%) compared to term neonates (45.2%). The difference was statistically significant, emphasizing that preterm neonates are inherently at greater risk of developing phototherapy-induced hypocalcemia. The higher susceptibility in preterm neonates can be attributed to immature parathyroid function and reduced calcium stores, which impair calcium homeostasis during stress conditions like phototherapy.

Similar findings were reported by Tan et al. (2019), who noted that preterm neonates are more likely to experience hypocalcemia due to impaired parathyroid hormone secretion and underdeveloped renal calcium reabsorption mechanisms (90). The study by Thakkar et al. (2020) also highlighted that low gestational age is a major risk factor for metabolic disturbances, including hypocalcemia, during phototherapy. These results advocate for more vigilant monitoring of serum calcium levels in preterm infants, especially when subjected to prolonged or intensive phototherapy.(91)

Distribution of Hypocalcemia by Neonatal Age (Days)

The distribution of hypocalcemia by neonatal age shows that the majority of hypocalcemic cases occurred on the first day of life (57.1%), followed by the second day (33.3%), and the

third day (9.5%). Despite this distribution, there was no statistically significant difference between the age groups, indicating that the age of presentation does not significantly influence the development of hypocalcemia following phototherapy.

The findings are consistent with those of Chang et al. (2020), who observed that hypocalcemia can develop regardless of the day of presentation, as long as the neonate undergoes phototherapy. This reflects that phototherapy-induced calcium depletion can occur irrespective of the neonate's age, emphasizing that metabolic disturbances are more closely related to the phototherapy process itself rather than the day of life.(73)

Studies by Mishra et al. (2018) also noted that early initiation of phototherapy does not necessarily reduce the risk of hypocalcemia, as the duration and intensity of phototherapy are more critical determinants. Therefore, irrespective of the age at presentation, continuous monitoring of serum calcium levels remains crucial during phototherapy.

Distribution of Hypocalcemia by Type of Phototherapy

The study observed a significantly higher incidence of hypocalcemia among neonates receiving intensive phototherapy (59.5%) compared to conventional phototherapy (40.5%). The difference was statistically significant, indicating that higher light intensity and longer exposure times in intensive phototherapy may lead to greater calcium depletion.

This finding is in line with the results reported by Romagnoli et al. (2019) and Murmu et al. (2024), who demonstrated that intensive phototherapy significantly impacts serum calcium levels due to increased light exposure. Intensive phototherapy, typically delivered at wavelengths around 460–490 nm, is known to suppress melatonin secretion, thereby reducing parathyroid hormone levels and impairing calcium mobilization from bones. This physiological mechanism explains the higher incidence of hypocalcemia observed in the intensive phototherapy group (86, 1).

To mitigate this risk, it is recommended to routinely monitor calcium levels in neonates undergoing intensive phototherapy and consider prophylactic calcium supplementation, particularly in preterm and low birth weight infants.

Symptoms of Hypocalcemia in Affected Neonates

The study found that 35.7% of hypocalcemic neonates exhibited symptoms such as jitteriness, poor feeding, irritability, or seizures, while the remaining 64.3% remained asymptomatic. This indicates that a significant proportion of neonates with phototherapy-induced hypocalcemia are asymptomatic, highlighting the importance of routine biochemical monitoring even in the absence of clinical symptoms.

This finding is consistent with studies by Arora et al. (2018) and Romagnoli et al. (2019), who reported that asymptomatic hypocalcemia is common during phototherapy, particularly in preterm and low birth weight neonates. They emphasized that reliance on clinical signs alone can lead to underdiagnosis and that laboratory confirmation is crucial (88, 86).

The presence of symptomatic hypocalcemia in over one-third of affected neonates in our study underscores the potential severity of phototherapy-induced calcium depletion. Symptomatic cases often require prompt intervention, including oral or intravenous calcium supplementation, to mitigate the risk of neurological complications.

Weight Distribution in Hypocalcemic and Non-Hypocalcemic Neonates

The mean birth weight of hypocalcemic neonates was significantly lower (2.0 kg) compared to non-hypocalcemic neonates (2.4 kg). This statistically significant difference underscores the role of low birth weight as a major risk factor for the development of hypocalcemia during phototherapy.

Several studies, including those by Narang et al. (2020) and Murmu et al. (2024), have demonstrated that low birth weight neonates are more vulnerable to metabolic disturbances, including hypocalcemia, due to limited calcium stores and immature parathyroid gland function (89, 1). Additionally, Thakkar et al. (2020) noted that the reduced bone mineral content in low birth weight infants makes them prone to calcium depletion, especially under the stress of phototherapy.(91)

The findings suggest that low birth weight should be considered a significant predictor of hypocalcemia risk, necessitating more frequent monitoring and possibly prophylactic calcium supplementation during phototherapy. Early identification and correction of hypocalcemia can prevent adverse neurological outcomes.

Serum Bilirubin Levels in Hypocalcemic and Non-Hypocalcemic Neonates

The study showed that hypocalcemic neonates had lower initial serum bilirubin levels (10.7 mg/dL) compared to non-hypocalcemic neonates (12.8 mg/dL). After 24 hours of phototherapy, bilirubin levels dropped more significantly in hypocalcemic neonates (4.9 mg/dL) compared to non-hypocalcemic ones (5.9 mg/dL).

This rapid decline in bilirubin levels among hypocalcemic neonates suggests that the rate of bilirubin reduction may be associated with calcium depletion. Studies by Maisels and Watchko (2019) indicated that intensive and effective phototherapy could lead to rapid bilirubin breakdown, potentially disrupting metabolic equilibrium and calcium homeostasis (57). Additionally, Romagnoli et al. (2019) noted that the photochemical conversion of bilirubin can interfere with calcium metabolism, particularly when phototherapy intensity is high.(86)

The correlation between more rapid bilirubin reduction and the development of hypocalcemia underscores the importance of balancing phototherapy efficacy with metabolic monitoring. It suggests that neonates with rapid bilirubin decline may require more stringent calcium level assessments to prevent metabolic disturbances.

Serum Calcium Levels in Hypocalcemic and Non-Hypocalcemic Neonates

The study found that the mean serum calcium level at the start of phototherapy was significantly lower in hypocalcemic neonates (8.4 mg/dL) compared to non-hypocalcemic neonates (9.1 mg/dL). After 24 hours of phototherapy, the calcium levels dropped further to 7.0 mg/dL in hypocalcemic neonates, while non-hypocalcemic neonates had a mean level of 8.5 mg/dL. The difference between the two groups was statistically significant ($p < 0.001$).

This finding highlights that neonates with lower baseline calcium levels are more susceptible to phototherapy-induced hypocalcemia. Studies by Romagnoli et al. (2019) and Arora et al. (2020) reported similar trends, emphasizing that neonates who start phototherapy with marginal calcium levels are more prone to significant declines during treatment. The proposed mechanism involves melatonin suppression and parathyroid hormone (PTH) reduction, resulting in decreased calcium mobilization from bone and increased urinary calcium loss.(86, 88)

Given these findings, it is crucial to identify neonates with borderline low calcium levels before initiating phototherapy and consider prophylactic calcium supplementation to mitigate

the risk of hypocalcemia. Close monitoring is particularly important in preterm and low birth weight neonates, where baseline hypocalcemia is more prevalent.

Correlation Between Type of Phototherapy and Hypocalcemia in Preterm and Term Neonates

The study revealed that intensive phototherapy significantly increased the risk of hypocalcemia, especially among term neonates (94.7%) compared to preterm neonates (30.4%). Conventional phototherapy posed a relatively lower risk, particularly in term neonates (5.3%). The significant p-values across all comparisons indicate that **higher phototherapy intensity correlates with greater calcium depletion**, regardless of gestational age.

This observation is consistent with studies by Tan et al. (2019) and Narang et al. (2020), who found that intensive phototherapy, with its higher irradiance and prolonged exposure, leads to more pronounced metabolic disturbances. The mechanism likely involves enhanced suppression of melatonin and parathyroid activity, impairing calcium homeostasis and increasing the risk of hypocalcemia.(90, 89)

The finding that term neonates are more affected than preterm neonates during intensive phototherapy may be explained by differences in metabolic adaptation and bone mineralization between the two groups. While preterm neonates generally have immature calcium regulation, term neonates may experience a sudden metabolic challenge when exposed to intensive phototherapy, leading to acute hypocalcemia. Therefore, intensive phototherapy should be used cautiously, with routine calcium monitoring, particularly in term neonates.

Correlation Between Phototherapy Type and Symptomatic Hypocalcemia in Preterm and Term Neonates

The study revealed that symptomatic hypocalcemia was significantly more common in neonates receiving intensive phototherapy, especially among term neonates (85.7%) compared to preterm neonates (37.5%). In contrast, conventional phototherapy was associated with a much lower incidence of symptomatic hypocalcemia. The significant p-values indicate that intensive phototherapy not only increases the risk of hypocalcemia but

also the likelihood of developing symptoms, such as jitteriness, poor feeding, irritability, and seizures.

These results are consistent with those of Murmu et al. (2024), who observed that intensive phototherapy significantly increases the incidence of symptomatic hypocalcemia, especially in neonates with higher birth weights and more developed metabolic systems (1). Another study by Bhutani et al. (2020) highlighted that symptomatic hypocalcemia is more frequently seen in term neonates undergoing high-intensity phototherapy, likely due to greater suppression of PTH activity and melatonin levels.(82)

The high rate of symptomatic hypocalcemia in term neonates receiving intensive phototherapy underscores the need for proactive management, including baseline calcium measurement and supplementation when needed. Additionally, shortening the duration of intensive phototherapy or transitioning to conventional methods when feasible may help mitigate the risk of symptomatic hypocalcemia.

CHAPTER VII

SUMMARY

- Term neonates (63%) were more prevalent than preterm neonates (37%), indicating a higher representation of mature infants.
- The mean birth weight was 2.2 kg, indicating a significant proportion of low birth weight neonates prone to hyperbilirubinemia.
- There was no significant difference between genders, with 55% females and 45% males requiring phototherapy.
- The majority of neonates (58%) presented for phototherapy on the first day of life, indicating early intervention for hyperbilirubinemia.
- Normal vaginal delivery (53%) was the most common mode of delivery, followed by cesarean section (41%).
- Conventional phototherapy (71%) was more commonly used compared to intensive phototherapy (29%).
- Serum bilirubin levels significantly decreased from 11.9 mg/dL to 5.5 mg/dL after 24 hours of phototherapy.
- Serum calcium levels significantly decreased from 8.8 mg/dL to 7.9 mg/dL after 24 hours of phototherapy, indicating hypocalcemia risk.
- Hypocalcemia was present in 42% of neonates, highlighting the need for calcium monitoring during phototherapy.
- Only 15% of hypocalcemic neonates were symptomatic, indicating the predominance of subclinical hypocalcemia.
- Preterm neonates (54.8%) were significantly more prone to hypocalcemia compared to term neonates (45.2%).
- Hypocalcemia incidence did not significantly differ by age, with most cases presenting on the first and second days of life.
- Intensive phototherapy was significantly associated with higher hypocalcemia incidence (59.5%) compared to conventional phototherapy (40.5%).
- Symptomatic hypocalcemia was present in 35.7% of affected neonates, while 64.3% remained asymptomatic.
- Hypocalcemic neonates had a significantly lower mean birth weight (2.0 kg) compared to non-hypocalcemic neonates (2.4 kg).

- Hypocalcemic neonates had significantly lower initial bilirubin levels (10.7 mg/dL) compared to non-hypocalcemic neonates (12.8 mg/dL).
- Serum calcium levels significantly dropped in hypocalcemic neonates (from 8.4 to 7.0 mg/dL) compared to non-hypocalcemic neonates.
- Intensive phototherapy significantly increased the risk of hypocalcemia, especially in term neonates (94.7%).
- Symptomatic hypocalcemia was significantly more common with intensive phototherapy, particularly among term neonates (85.7%).

CHAPTER VIII

CONCLUSION

Phototherapy remains an essential and effective treatment for neonatal hyperbilirubinemia, significantly reducing serum bilirubin levels and preventing bilirubin-induced neurological dysfunction. However, this study highlights that phototherapy-induced hypocalcemia is a considerable metabolic complication, affecting 42% of neonates undergoing treatment. Intensive phototherapy, in particular, is associated with a higher risk of hypocalcemia compared to conventional phototherapy, especially among term neonates.

Preterm neonates and those with low birth weight are particularly vulnerable to developing hypocalcemia due to immature calcium homeostasis and reduced calcium stores. The significant drop in serum calcium levels observed after 24 hours of phototherapy indicates that even asymptomatic hypocalcemia may go undetected without routine biochemical monitoring. The fact that only 15% of hypocalcemic neonates exhibited clinical symptoms underscores the importance of proactive screening rather than relying solely on clinical presentation.

The study also found that rapid reductions in serum bilirubin levels were associated with an increased risk of hypocalcemia, suggesting that intensive phototherapy may lead to more pronounced metabolic disturbances. Additionally, while gender and mode of delivery did not significantly impact hypocalcemia risk, intensive phototherapy and gestational maturity emerged as key determinants.

To minimize the risk of phototherapy-induced hypocalcemia, routine monitoring of serum calcium levels should be incorporated into neonatal care protocols, particularly for neonates undergoing intensive phototherapy. Early detection and prompt calcium supplementation can effectively mitigate the potential complications associated with hypocalcemia. Educating healthcare providers about the metabolic risks of phototherapy and implementing standardized monitoring practices can enhance patient safety and improve neonatal outcomes.

PROFORMA

Study Title:

Phototherapy-Induced Hypocalcemia in Term and Preterm Neonates with Neonatal Hyperbilirubinemia and Its Clinical Significance in a Tertiary Care Hospital of Southern Haryana

Patient Identification Details:

1. Name of the Neonate: _____
2. Hospital ID/Registration Number: _____
3. Date of Admission: _____
4. Date of Birth: _____
5. Gender: ☐ Male ☐ Female

Maternal Details:

6. Maternal Age (years): _____
7. Antenatal Complications (if any): _____
8. Maternal Blood Group:
☐ A+ ☐ A- ☐ B+ ☐ B- ☐ AB+ ☐ AB- ☐ O+ ☐ O-
9. History of Previous Neonatal Jaundice:
☐ Yes ☐ No
If yes, specify: _____

10. History of Calcium/Vitamin D Supplementation during Pregnancy:

☐ Yes ☐ No

If yes, specify dose and duration: _____

Birth and Delivery Details:

11. Gestational Age at Birth (weeks): _____

12. Birth Weight (kg): _____

13. Mode of Delivery:

- ☐ Normal Vaginal Delivery (NVD)
- ☐ Lower Segment Cesarean Section (LSCS)
- ☐ Forceps Delivery
- ☐ Vacuum Extraction

14. Apgar Score at 1 minute: _____

15. Apgar Score at 5 minutes: _____

Neonatal Details:

16. Baby's Blood Group:

☐ A+ ☐ A- ☐ B+ ☐ B- ☐ AB+ ☐ AB- ☐ O+ ☐ O-

17. Age at Presentation (days): _____

18. Clinical Symptoms (if any):

- ☐ Jitteriness
- ☐ Irritability
- ☐ Poor Feeding
- ☐ Seizures
- ☐ Apnea
- ☐ None

Phototherapy Details:

19. Type of Phototherapy Administered:

- ☐ Conventional Phototherapy
☐ Intensive Phototherapy

20. Duration of Phototherapy (hours): _____

21. Total Hours of Exposure: _____

22. Distance from Light Source (cm): _____

23. Intensity of Phototherapy (W/cm^2): _____

Laboratory Investigations:**Serum Bilirubin Levels:**

24. At Start of Phototherapy (mg/dL): _____

25. After 24 Hours of Phototherapy (mg/dL): _____

Serum Calcium Levels:

26. At Start of Phototherapy (mg/dL): _____

27. After 24 Hours of Phototherapy (mg/dL): _____

Other Investigations:

28. Direct Coombs Test (DCT): ☐ Positive ☐ Negative

29. Peripheral Smear for Hemolysis: ☐ Present ☐ Absent

30. Blood Glucose Level (mg/dL): _____

31. Serum Albumin Level (g/dL): _____

32. Serum Magnesium Level (mg/dL): _____

Neonatal Outcome:

33. Hypocalcemia Status:

- ☐ Absent (≥ 8 mg/dL for term; ≥ 7 mg/dL for preterm)
- ☐ Present (< 8 mg/dL for term; < 7 mg/dL for preterm)

34. Management of Hypocalcemia:

- ☐ Oral Calcium Supplementation
- ☐ Intravenous Calcium Gluconate
- ☐ None

35. Complications Observed:

- ☐ None
- ☐ Seizures
- ☐ Cardiac Arrhythmia
- ☐ Hypotonia
- ☐ Others (specify): _____

Follow-Up Details:

36. Post-Treatment Serum Calcium (mg/dL): _____

37. Post-Treatment Serum Bilirubin (mg/dL): _____

38. Discharge Status:

- ☐ Recovered
- ☐ Referred
- ☐ Deceased
- ☐ Discharged Against Medical Advice (DAMA)

Investigator's Signature: _____

Date: _____

REFERENCES